Clin Child Fam Psychol Rev (2012) 15:303–329
DOI 10.1007/s10567-012-0121-z

Developmental Context and Treatment Principles for ADHD Among College Students

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Published online: 29 September 2012

Abstract Attention-deficit/hyperactivity disorder (ADHD) affects between 2 and 8 % of college students. ADHD is associated with impaired academic, psychological, and social functioning, and with a wide array of negative outcomes including lower GPAs, graduation rates, and self-reported quality of life. The college environment often brings decreased external structure and increased availability of immediate rewards, presenting added demands for behavioral self-regulation—an area in which students with ADHD are already vulnerable. Despite the significant impact of ADHD in college and the unique challenges presented by the college context, virtually no treatment development research has been conducted with this population. In order to provide a framework to guide intervention development, this comprehensive review integrates research from three key domains that inform treatment for college students with ADHD: (1) functional impairment associated with ADHD among college students, (2) etiology of ADHD and the developmental context for ADHD among emerging adults (age 18–24), and (3) treatment outcome research for ADHD among adolescents and adults. A detailed set of proposed treatment targets and intervention principles are identified, and key challenges associated with treatment development in this population are discussed.

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Keywords ADHD · College · Treatment · Executive functioning · Development · Emerging adults

Introduction

Attention-deficit/hyperactivity disorder (ADHD)—once thought to be a disorder of childhood—persists into adulthood in up to 80 % of individuals diagnosed as children (Barkley et al. 2007). ADHD is characterized by developmentally atypical levels of inattentive behavior, hyperactive and impulsive behavior, or both. Among adults, ADHD is associated with significant adaptive impairment across several domains, including academic, occupational, social, and psychological functioning. Several psychosocial and psychopharmacological treatments have been developed to address ADHD among adults. Many of these interventions have shown efficacy in randomized controlled trials, including stimulant medication (Prince 2006), individual cognitive-behavioral therapy (CBT), and group CBT (Knouse et al. 2008). Other approaches, such as mindfulness training, have shown preliminary evidence of efficacy (Zylowska et al. 2008) but have not yet been evaluated in randomized trials. This burgeoning evidence base on treatments for adults with ADHD is advancing to meet the extensive literature on treatments for children and adolescents with ADHD that has been amassed over the past 40 years (Knight et al. 2008).

In contrast, the population of *emerging adults* (age 18–24) has received very little attention in treatment development research for ADHD. Because ADHD often impairs many of the behaviors essential for adaptive functioning as an adult in society (e.g., organization, long-term planning, mood regulation), developmental progress toward adaptive coping during this crucial period of transition from adolescence to adulthood may powerfully affect the long-term developmental trajectory for individuals with ADHD.

For many individuals, with or without ADHD, college is the dominant context in which the majority of this developmental period of emerging adulthood occurs. Between 2 and 8 % of college students meet criteria for ADHD (DuPaul et al. 2009). College students with ADHD tend to have higher cognitive abilities and more effective compensatory skills than their peers diagnosed with ADHD who do not attend college (Frazier et al. 2007); however, they also experience an environment that presents a unique combination of stressors and challenges for those who struggle with sustained attention, organization, self-management, and pursuit of long-term rewards over short-term rewards. ADHD-related deficits in these aforementioned areas combine with an abrupt loss of parental structure and support and a still-maturing neurological system supporting self-regulation to produce significant adaptive impairment for many college students with ADHD. Compared to their peers without ADHD, students with ADHD tend to have lower GPAs, higher rates of academic probation and dropout, and poorer functioning across several other domains (e.g., social, psychological) (e.g., Blase et al. 2009).

Despite the urgent need for effective, evidence-based interventions for ADHD among college students, very few reports of psychosocial interventions with this population have been published. This has two important implications: (1) treatment development research in this area is urgently needed and (2) no clear prescriptions for treating college ADHD can be drawn at this time. Nevertheless, treatment development research as well as ongoing clinical intervention efforts will benefit from guidance based upon the best available evidence.

The current review combines evidence from three key domains that inform treatment for college students with ADHD: (1) quantitative and qualitative data on functional impairment among college students with ADHD, (2) etiology of ADHD and the developmental context for ADHD among emerging adults, and college students in particular, and (3) treatment outcome research on adolescents and adults with ADHD. Based upon a synthesis of research evidence in each of these three areas, we present a detailed list of proposed targets for intervention and corresponding treatment principles tailored to the unique presentation of ADHD in college. Although tentative, these proposed targets are intended to serve a heuristic function for treatment and research. Subsequently, several key issues related to treatment development work with this population are discussed.

Functional Impairment Associated with College ADHD

A growing body of research describes the broad-ranging impairment associated with ADHD in college. The following section provides a brief overview of the evidence on ADHD-related functional impairment in college, with an emphasis on identifying behavioral targets for intervention. Two recent reviews (DuPaul et al. 2009; Weyandt and DuPaul 2008) provide more extensive detail and discussion of prevalence, symptomatology, and assessment of ADHD among college students.

Academic Functioning

Research has consistently shown that college students with ADHD tend to have lower academic performance than their peers without ADHD on a variety of measures.

Several studies have found that students with ADHD have lower GPAs than students without ADHD, both in the first year of college (Frazier et al. 2007; Glutting et al. 2002; Rabiner et al. 2008) and throughout college (Blase et al. 2009; Heiligenstein et al. 1999). Students with ADHD are more likely than non-ADHD peers to be placed on academic probation (Frazier et al. 2007; Heiligenstein et al. 1999) and less likely to graduate from college (Heiligenstein et al. 1999; Wolf 2001). Similarly, students with ADHD have higher self-reported academic problems (Heiligenstein et al. 1999; Lewandowski et al. 2008). Among normative samples of college students, greater self-reported inattentive symptoms predicted lower self-reported academic adjustment and career decision-making self-efficacy (Norwalk et al. 2009), and lower GPAs (Schwanz et al. 2007). Cross-cultural research has found that academic impairment (as well as social and career-related impairment) due to ADHD is not specific to American college students, but also occurs among Chinese undergraduates (Norvilitis et al. 2010).

Nearly all studies that have independently examined inattentive and hyperactive/impulsive symptom clusters have found that inattentive behaviors (either self-reported or parent-reported) predict academic impairment, while hyperactive/impulsive behaviors do not (e.g., Frazier et al. 2007; Glutting et al. 2002; Rabiner et al. 2008). These findings parallel those of the adult ADHD literature (e.g., Stavro et al. 2007) suggesting that inattentive symptoms predict the majority of variance in adaptive functioning. Although further study is needed to clarify the precise nature of this relationship, it is suggested that inattentive ADHD symptoms interfere with key behaviors leading to academic success in college—in particular, the capacity to sustain attention for long periods of time, organize complex tasks, and independently initiate and complete tasks.

Several proximal academic-related behaviors have been shown to be associated with ADHD. Although not yet empirically tested, these behaviors may mediate the relation between ADHD and impaired academic functioning. Multiple studies have found that, relative to their peers without ADHD, college students with ADHD show poorer organization and time management skills (Kern et al. 1999), more procrastination behavior (Turnock et al. 1998), and lower-quality writing independent of spelling, punctuation, grammar, and handwriting (Gregg et al. 2002). College students with ADHD also show poorer performance on timed tests, difficulty completing tests on time, longer studying time to complete assignments, and greater perception of having to work harder to achieve comparable grades as non-ADHD peers (Lewandowski et al. 2008).

When compared to both normal controls and students with learning disabilities, college students with ADHD show lower scores on four domains of the learning and study strategies inventory (LASSI; Weinstein et al. 2002): time management, concentration, selection of main ideas, and test-taking strategies (Reaser et al. 2007). Relative to normal controls, both the ADHD and LD groups also scored lower on motivation, anxiety, information processing, and self-testing. Researchers have suggested that ADHD-related academic impairment may spike with the transition to college due to the loss of structure and support of the family environment (e.g., Heiligenstein et al. 1999; Meaux et al. 2009).

Psychological Functioning

College students with ADHD tend to show poorer psychological adjustment and lower self-reported quality of life than their peers without ADHD (Grenwald-Mayes 2001).

Numerous studies have found that ADHD among college students predicts higher levels of depressive symptoms (e.g., Blase et al. 2009; Heiligenstein and Keeling 1995; Rabiner et al. 2008), although one study failed to replicate this finding (Heiligenstein et al. 1999). Similarly, students with ADHD show higher rates of overall psychological distress (Richards et al. 1999; Weyandt et al. 1998) on broad symptom measures such as the Symptom Checklist-90 (SCL-90; Derogatis 1975) and the Brief Symptom Inventory (BSI; Derogatis 1992).

Difficulty regulating emotions has been consistently found to be associated with ADHD among children, adolescents, and adults (Barkley 2006). Adults with ADHD show more state and trait anger, and greater difficulty expressing anger in a socially acceptable manner (Ramirez et al. 1997). Research suggests that, among individuals with ADHD, symptoms of depression and anxiety increase throughout the transition to adulthood as adaptive functioning expectations increase and ADHD-related behaviors contribute to experiences of failure in school, work, and relationships (e.g., Safren et al. 2010a). Among college students in particular, ADHD appears to be associated with more difficulties in monitoring personal stress and in accessing social support from others (Kern et al. 1999).

College students with ADHD are also at higher risk for substance use and abuse. Among a large sample of undergraduate students, ADHD status at college entry predicted smoking initiation and alcohol use (Blase et al. 2009; Rabiner et al. 2008). Even after controlling for history of conduct disorder, college students with ADHD show higher rates of tobacco use, dangerous alcohol use, negative consequences of alcohol use (e.g., having a hangover, getting into an argument or fight, experiencing memory loss) and impairment related to illicit drug use (Rooney et al. 2012). Notably, students with ADHD do not appear to consume alcohol in greater quantities or more frequently than peers without ADHD, but rather appear more vulnerable to developing dependence or other negative consequences of use (Smith et al. 2002). These results parallel findings that individuals with ADHD are more vulnerable to developing dependence on nicotine and other drugs due to their effect on the dopamine-modulated reward system, a primary neural circuit underlying ADHD behavior (Wilens et al. 2008).

Social Functioning

College students with ADHD report lower social skills and social adjustment than peers matched on age, gender, and GPA (Shaw-Zirt et al. 2005), with moderate-to-large effect sizes. Impaired social functioning appears to manifest in several specific areas.

Inattentive symptoms among male college students have been found to be associated with delayed dating onset, lower comfort and assertiveness in heterosocial situations, and lower ratings from female confederates following 1-min interactions (Canu and Carlson 2003). In qualitative interviews, female students frequently described themselves as blunt, and that they often blurted out comments that were hurtful to others (Meaux et al. 2009). Based on assessments of personality characteristics, individuals with ADHD appear to be more “confrontive” and “aggressive,” more independent, less rule-focused, and less influenced by corrective feedback, whether positive or negative (Kern et al. 1999). Based on retrospective reports, college students with ADHD also report poorer parent–child relations than their peers without ADHD (Grenwald-Mayes 2001).

College students with and without ADHD tend to endorse more negative adjectives than positive adjectives to describe a college student with ADHD (Chew et al. 2009). Those with ADHD actually endorsed more negative attitudes toward individuals with ADHD than did students without ADHD. Notably, more frequent contact with individuals with ADHD was associated with more favorable ratings, suggesting the importance of facilitating peer normalization in intervention for college students with ADHD.

Among a sample of first-semester college freshman, no differences in social satisfaction were found between students with and without ADHD (Rabiner et al. 2008). In combination with other findings, these results suggest that social dissatisfaction may grow over time as negative social experiences accrue. Therefore, early intervention may help to prevent development of greater social impairment throughout college.

Occupational Functioning

College students with ADHD report lower confidence in their own career decision-making ability (Career Decision-Making Self-Efficacy scale [CDMSE]; Betz 2000), a measure that is related to academic and social adjustment. These results were replicated in a recent study by Norwalk et al. (2009).

College students with ADHD show lower self-reported job performance and report having been fired more frequently than students without ADHD (Shifrin et al. 2010). Areas of occupational impairment endorsed most commonly were similar to those reported in the academic setting: (1) performance of assigned work, (2) management of daily responsibilities, and (3) management of time and work-related deadlines. These findings suggest that interventions that effectively target students' academic functioning may also serve to address important deficits in occupational functioning.

Driving Performance

ADHD among college students is also associated with more overall driving citations, speeding violations, license suspensions and revocations, and motor vehicle accidents (Barkley et al. 2002), as well as more anger while driving and high rates of aggressive or risky driving (Richards et al. 2002). Similarly, ADHD symptoms at age 13 predict a poorer driving record at age 21, including more motor vehicle accidents, traffic violations, and arrests for drinking and driving (Woodward et al. 2000).

Impairment into Adulthood

Several studies have demonstrated that impairment associated with ADHD often continues into adulthood. Similarly to ADHD among college students, ADHD among adults is linked to poorer outcomes across a broad range of domains, including occupational, interpersonal, and psychological functioning (e.g., Biederman et al. 2006a; Brod et al. 2005). Adults with ADHD also report lower quality of life than those without ADHD (Brod et al. 2006).

Among adults with ADHD between ages of 18 and 65, overall severity of ADHD symptoms was found to uniquely predict impairments in occupational and interpersonal functioning (Safren et al. 2010a). Comorbid anxiety and depressive symptoms also independently predict poorer interpersonal functioning and lower life satisfaction.

Adults with ADHD experience greater difficulty with employment stability based on several metrics, including poorer employer-rated work performance, shorter employment duration, and higher rates of impulsively quitting or being involuntarily terminated from jobs (e.g., Barkley 2002; Murphy and Barkley 1996). Similar occupational impairments were found among a large sample ($N = 414$) of adults with ADHD located in Norway (Halmøy et al. 2009). Far fewer individuals with ADHD (24 %) than normative control subjects (79 %) were currently employed. Among those with ADHD, the strongest predictor of employment in adulthood was previous treatment with stimulant medication during childhood (odds ratio = 3.2). Although interpretation is limited by the correlational design, these results point to the developmental nature of ADHD-related impairment and suggest that effective treatment for ADHD prior to adulthood may have significant positive long-term implications.

Etiology of ADHD

ADHD behaviors are the product of a complex transaction between an individual and his or her environment over time. Most individuals with ADHD carry a strong genetic vulnerability factor. The heritability (or proportion of trait variation that is due to genetic variation) for ADHD is estimated at 0.77 (Faraone and Doyle 2001; Wallis et al. 2008), which is highest among all psychiatric disorders, and approaches that of adult height. In utero exposure to teratogens (e.g., lead, nicotine) is also linked to ADHD (Barkley 2006). Biological vulnerabilities to ADHD operate through certain key neurological systems, particularly those served by the neurotransmitters dopamine and norepinephrine (Vaidya and Stollstorff 2008). Over time, these biological factors transact with the environment to result in ADHD criterion behavior. Although the vast majority of etiological research on ADHD focuses on children and adolescents, the conceptualization of ADHD as a developmental disorder with a coherent trajectory suggests that the extant literature may be directly applied to understanding the nature of ADHD among college students and adults (Barkley et al. 2007).

Research over the past 10 years strongly suggests a heterogeneous etiology for ADHD. Evidence supports the assertion that ADHD-related impairment results from deficits in one or more of several neurological processes, including "cool" executive functions (e.g., response inhibition and working memory), "hot" executive functions (motivation toward delayed vs. immediate rewards), and temporal processing. Although these areas interact, impairment in each area appears to be dissociable from the others. The following section provides a brief overview of evidence supporting each of these etiological pathways to ADHD symptomatology.

Response Inhibition and Working Memory ("Cool" Executive Functioning)

For many individuals with ADHD, functional impairment may arise from a core deficit in response inhibition (Barkley 1997), which may also give rise to secondary deficits in verbal and non-verbal working memory, self-regulation of emotion, and planning (analysis and synthesis). Response inhibition (also called behavioral inhibition) includes (1) inhibiting a "prepotent response" (or automatic response) to an event, (2) stopping an ongoing response in favor of a more adaptive response, and (3) protecting an ongoing response from competing or conflicting responses (interference control) (Barkley 2006). Working memory permits a limited amount of information to be held "on-line" for use by the central executive processes and includes both verbal/auditory and non-verbal/visuospatial components (Baddeley and Hitch 1974).

Neuroimaging research supports the claim that deficits in "cool" executive functions are involved in ADHD. Response inhibition and working memory are supported primarily by the frontostriatal neural circuit, comprised of the lateral prefrontal cortex, dorsal anterior cingulate, and dorsal striatum (especially the caudate), with links to the cerebellum through the thalamus. Structural neuroimaging found that individuals with ADHD show cortical thinning in the dorsolateral prefrontal and anterior cingulate cortices (Makris et al. 2007). A meta-analysis of neuroimaging studies found consistent underactivation in the frontostriatal circuit among subjects with ADHD during tasks that require response inhibition, maintenance and manipulation of working memory, monitoring of errors, and interference control (Dickstein et al. 2006).

Behavioral studies also implicate "cool" executive function deficits in ADHD. Results from neuropsychological tests designed to tap executive functions show that individuals with ADHD tend to perform more poorly than normal controls. Large meta-analyses of studies assessing children and adolescents (Willcutt et al. 2005) and adults (Boonstra et al. 2005; Hervey et al. 2004) have shown moderate-to-large effect size differences in response inhibition, working memory, planning, and attention. Performance decrements tend to increase with the cognitive demands of the task (e.g., complexity, time requirements, processing speed demand).

Longitudinal research has shown that executive function deficits associated with ADHD are fairly stable throughout the lifespan (Biederman et al. 2007; Loo et al. 2007; Murphy et al. 2001). These findings suggest that the executive function impairment associated with ADHD is not merely the result of *delayed* development, but rather of a fundamental deficit that is consistent over time in most individuals. In fact, some evidence points to larger executive deficits among adults than among children (Lijffijt et al. 2005). Furthermore, executive function deficits are associated with poorer functioning across several domains, including lower academic achievement and socioeconomic status (Biederman et al. 2006b; Loo et al. 2007).

Assessments of executive functioning among college students have shown mixed results. Gropper and Tannock (2009) found that students with ADHD show poorer working memory performance than normal controls. Notably, poorer auditory-verbal working memory ability was specifically associated with lower GPAs. Conversely, other studies have found limited ADHD-related differences in response inhibition (Weyandt et al. 2002) and have failed to detect differences in divided attention (Linterman and Weyandt 2001) or conflict attention (Weyandt et al. 1995). Taken together, these findings suggest that "cool" executive function deficits contribute to ADHD-related impairment in some, but not all, college students.

Delay Aversion ("Hot" Executive Functioning)

Deficits in responding to delayed rewards have also been clearly linked to ADHD. Greater delay aversion (the preference for smaller, immediate rewards over larger, delayed rewards) appears to underlie a significant portion of ADHD behavior (Sonuga-Barke 2005). In such cases, impaired signaling of delayed rewards (Sagvolden et al. 2005) appears to produce a greater negative emotional reaction to the imposition of a delay, resulting in lower motivation toward long-term rewards. Thus, whereas deficits in "cool" executive function result in difficulty inhibiting an automatic action, deficits in "hot" executive functions result in greater preference for immediate, smaller rewards over larger, delayed rewards. Even if effective response inhibition allows for an intentional choice to be made, individuals with greater delay aversion are more likely to engage in behavior that is immediately reinforced. Impaired "hot" executive processes may also manifest as a deficit in extinction learning (i.e., neural signaling that a previously available reward is no longer available); thus, individuals with ADHD may persist longer in a behavior that has ceased to be functional.

Neuroimaging research indicates that "hot" executive functions are underpinned by the mesolimbic circuit. The mesolimbic circuit connects ventromedial prefrontal regions, including the orbitofrontal gyri and anterior cingulate, with the ventral striatum (nucleus accumbens), regions of the hippocampus and amygdala, and the ventral tegmental area. Individuals with ADHD tend to show atypical physiological response to reward as well as greater delay aversion and more risky decision-making than normal controls (Luman et al. 2005). Functional neuroimaging research has shown that both children (Scheres et al. 2007) and adults (Strohle et al. 2008) with ADHD have lower activation in the ventral striatum during anticipation of reward, but higher activation upon the delivery of reward. Individuals with ADHD also show cortical thinning in the anterior cingulate cortex (Makris et al. 2007) and lower "top-down" inhibitory control from the orbitofrontal areas to the ventral striatum (Strohle et al. 2008).

Behavioral research has consistently found ADHD-related differences in performance on tasks assessing delay aversion. Youth with ADHD and their siblings were both more likely to choose smaller, immediate rewards than normal controls (Marco et al. 2009). Boys with ADHD (10–18 years of age) show greater delay aversion than boys with OCD and normal controls (Vloet et al. 2010). Other studies have shown similar deficits in "hot" executive functioning among many individuals with ADHD (e.g., Solanto et al. 2001; Sonuga-Barke 2005). Additionally, treatment with stimulant medication appears to reduce deficits in response to delayed reward (Shiels et al. 2009). Further research is needed to clarify the connection between delay aversion and ADHD-related functional impairment among emerging adults and adults.

Temporal Processing

ADHD-related impairment may also result from deficits in processing of time. Children with ADHD tend to perform more poorly on time estimation tasks (Castellanos and Tannock 2002). Many individuals who meet criteria for ADHD perform in the normal range on tests of response inhibition and delay aversion, yet show impairment on temporal processing tasks such as tapping at a steady pace, discriminating between shorter and longer brief time intervals, and anticipating the end of a brief, consistent time interval (Sonuga-Barke et al. 2010). Neuroimaging research suggests that impaired temporal processing is associated with reduced activity in the cerebellum (Durston et al. 2007); however, these studies have not dissociated temporal processing from "cool" executive functions, which also involve the cerebellum (Vaidya and Stollstorff 2008).

Cognitive-Energetic State

Impairment on response inhibition or other executive functioning tasks may be due, at least in part, to deficits in arousal, activation, and effort—processes that control the *allocation* of cognitive resources—rather than impaired cognitive resources themselves (Sergeant 2000, 2005). One proposed measure of this type of state regulation difficulty is reaction time (RT) variability (consistency of speed of a response following a stimulus). RT variability is one of the most common deficits in ADHD (Castellanos and Tannock 2002). On average, children with ADHD show greater RT variability across a wide range of executive function tasks (e.g., Attentional Network Task, Stop Signal Test) (Epstein et al. 2011). This effect appears to be independent of task factors (i.e., rate of event presentation) and ADHD subtype. However, only about half of individuals with ADHD fall into the impaired range on RT variability measures (Nigg et al. 2005), and, independent of RT variability, task accuracy appears to contribute to differences in performance between those with and without ADHD (Epstein et al. 2011). These results further support the consistent finding that there are multiple pathways to ADHD symptomatology and impairment.

Neuropathological Heterogeneity (Multiple Pathways) and Key Implications

Accumulating research evidence supports the assertion that there are multiple developmental pathways to ADHD. Several behavioral studies have demonstrated that deficits in response inhibition and working memory, response to delayed reward, and temporal processing manifest independently (Lambek et al. 2010; Sonuga-Barke et al. 2010). In a sample of children with ADHD, measures of response inhibition (e.g., Stop Signal Test) and delay aversion (e.g., choice-delay task [C-DT]) were uncorrelated (Solanto et al. 2001). Furthermore, while each of these two tasks had only modest discriminative validity when used alone, a combination of the two correctly identified 87.5 % of individuals as ADHD or non-ADHD (sensitivity: 89.53 %; specificity: 85 %). Similarly, Sonuga-Barke et al. (1994) found that children with ADHD can often wait for rewards even when response inhibition is required and often choose not to wait even when response inhibition is not required. Response inhibition and delay aversion appear to operate independently among adolescents and young adults as well (Steinberg et al. 2009).

Neuroimaging research shows that neurological systems underlying "hot" and "cool" executive functions are related, yet dissociable (Sonuga-Barke et al. 2003). Furthermore, comparisons between individuals with ADHD and normal controls show very large effect sizes of differences in ADHD symptoms, while effect sizes of differences in executive functioning are smaller (in the moderate-to-large range). Only about 50 % of adolescents and adults with ADHD show impaired performance on tests of response inhibition (Nigg et al. 2005). Thus, it is unlikely that one pathway accounts for all cases of ADHD.

This neurological heterogeneity of ADHD has important implications for intervention. ADHD must not be viewed as a monolithic disorder for which the same intervention techniques will be effective with each individual. Some individuals may struggle primarily with organization and planning; others may have continual difficulty sustaining attention or working for long-term rewards; still others may run into trouble with awareness of time or with inconsistency in arousal and effort. Assessment and treatment for ADHD must seek to identify the core deficits for each individual and tailor interventions accordingly.

Inattentive Symptoms and Functional Impairment

Notably, as described in the previous section, inattentive symptoms appear to be the primary driver of functional impairment among both college students and adults with ADHD. In a study comparing adults (ages 18–37) with and without ADHD, inattentive-disorganized symptoms accounted for 67.2 % of the variance in adaptive functioning while hyperactive-impulsive symptoms accounted for only 3.6 %. Several studies of college students have shown similar associations between greater inattentive symptoms and poorer academic, psychological, and career-related outcomes (Betz 2000; Blase et al. 2009; Frazier et al. 2007; Norvilitis et al. 2010; Norwalk et al. 2009; Rabiner et al. 2008; Schwanz et al. 2007). Deficits in both response inhibition (Stavro et al. 2007) and working memory (Gropper and Tannock 2009; Loo et al. 2007) have been linked to adaptive impairment. No studies have examined associations between measures of delay aversion or temporal processing and functional outcomes; further research in this area is needed.

Given the consistency and strength of empirical data implicating inattentive-disorganized symptoms as the key driver of functional impairment among college students with ADHD, interventions should primarily target inattentive-disorganized behavior rather than hyperactive-impulsive behavior.

Developmental Context for ADHD in College

The previous section provided a general review on neurological mechanisms underlying ADHD behavior. In order to maximize the usefulness of this research for informing treatment of ADHD among college students, it is important to understand how the unique developmental context of emerging adulthood and the environmental context of college affect the presentation of ADHD. The transition from adolescence to adulthood brings important neurobiological changes as well as dramatic sociocultural changes.

Executive Function Development

The neural structures underlying executive functioning are among the latest to mature, continuing to develop throughout the mid-20s (Giedd 2004). As a result, typical college students begin college—and in most cases, graduate from college—with executive systems that are still "under construction." A key process in the maturation of all neural circuits is myelination, or the encasing of neural axons within a fatty sheath. Myelinated axons transmit neural impulses up to 100 times faster; the dramatic increase in neural efficiency afforded by myelination facilitates complex cognitive processing and integration of inputs from multiple sources. Until this process of myelination is complete, emerging adults are more vulnerable to errors in self-regulation, especially when the task demands are high (e.g., complex tasks, high emotional arousal) (Steinberg 2007).

"Cool" Executive Functions

Anatomical magnetic resonance imaging (MRI) results indicate that myelination of the dorsal lateral prefrontal cortex (DLPFC) continues throughout the early twenties (Giedd 2004). As described above, the DLPFC is particularly involved in response inhibition and planning—core areas of executive functioning deficit among many individuals with ADHD. Diffusion tensor imaging (DTI) data showed that myelination of the frontostriatal circuit predicted faster reaction times on a response inhibition task (Go/No-Go)—with age and accuracy controlled—among individuals between the ages of 7 and 31 (Liston et al. 2006). This association was stronger when the task parameters required greater cognitive control. Similarly, performance on tasks measuring interference control, set shifting, and working memory continued to improve through age 21 (Huizinga et al. 2006).

"Hot" Executive Functions

The development of response to long-term reward is less well researched than that of response inhibition. Neuroimaging research suggests that adolescents show greater activity than adults in the nucleus accumbens (a key part of the mesolimbic circuit described above) in response to reward (e.g., Ernst et al. 2005); however, no known studies have provided a more fine-grained assessment of development through emerging adulthood. Similarly, we are aware of no studies that have used the C-DT behavioral paradigm—the task that has yielded the strongest empirical support for the importance of delay aversion in ADHD—to examine development of delay aversion into emerging adulthood.

Data from questionnaire-based investigations suggest that development of long-term reward motivation may continue throughout emerging adulthood. Among a large sample of adolescents, emerging adults, and adults, self-reported orientation toward future rewards (including items such as, “I often do things that don’t pay off right away but will help in the long run”) increased throughout adolescence into young adulthood (Cauffman and Steinberg 2000). In another similar study, self-reported planning and anticipation of future consequences did not reach its peak until ages 22–25 (Steinberg et al. 2009).

Results from a computer-based monetary rewards task indicated that delay discounting (a similar construct to delay aversion) decreased until age 16, but did not change significantly thereafter (Steinberg et al. 2009). However, this task involved a cognitive appraisal of hypothetical immediate versus delayed rewards, rather than a behavioral choice (that is, subjects did not receive the immediate or delayed monetary reward). Thus, the motivational processes governing reward-seeking behavior may not have been activated as they would be under real-world conditions. Further behavioral and neuroimaging research using active reward contingencies is needed in order to better understand the maturation of delay aversion throughout emerging adulthood.

Power of Socio-Emotional Context

A growing body of evidence suggests that socio-emotional context plays a powerful role in decision-making processes among adolescents and emerging adults (e.g., Gardner and Steinberg 2005; Steinberg 2007, 2010). Although adolescents show adult-like ability to understand and evaluate risk and long-term consequences by around age 16 (e.g., Fischoff 1992), recent research shows that the social and emotional contexts under which *actual* decisions are made are often very different from controlled, laboratory settings.

Gardner and Steinberg (2005) used a computer-based driving program to measure reward-related decision-making among adolescents, emerging adults, and adults. Participants earned rewards by advancing the car farther along the road, but risked losing rewards if the car crashed. Results showed a significant interaction between age and peer presence. When playing the game with two peers present rather than alone, adolescents experienced more than double the number of crashes, and college undergraduates experienced 50 % more crashes. Conversely, the number of crashes among adults was unaffected by peer presence. Thus, it appears that the motivational system (reward-seeking) may overwhelm the executive function system under contexts of high socio-emotional arousal. This effect appears most pronounced among adolescents, yet remains powerful among emerging adults as well. The attenuation of this phenomenon by full-fledged adulthood may be explained by the ongoing myelination of neural pathways underlying and connecting the motivational and executive systems. As these neural structures become more efficient and coordinated, the ability to effectively regulate behavior in high-demand contexts increases.

These findings may be particularly relevant for college students with ADHD. As described above, these individuals likely experience a double-deficit (maturational and ADHD-related) in response inhibition and long-term reward motivation. Additionally, their ability to direct behavior toward longer-term rewards may be particularly compromised in situations where salience of immediate rewards is heightened by socio-emotional arousal. Given the high degree of social interaction in most college settings, the power of socio-emotional context is especially important to consider in designing interventions for college students with ADHD.

Emerging Adulthood, College, and the “Double-Deficit” in Self-Regulation

Emerging Adulthood

In the United States and many cultures around the world, the time from age 18 to 24 (emerging adulthood) is a critical period of developmental transition from child to adult roles in society. This period brings dramatic changes in lifestyle, independence, and responsibility, often including: moving apart from family, engaging in self-directed academic programs, working full-time, managing finances independently, pursuing a career, and navigating long-term romantic relationships. Furthermore, the stakes are high for fledgling adults: Adjustment during this crucial period of emerging adulthood often predicts adjustment much later into adulthood (Schulenberg et al. 2004).

College

For most students, the beginning of college brings an abrupt shift into emerging adulthood (Farrell 2003). Most students experience a marked loss of parental supervision and structure, and a corresponding dramatic increase in independence in academic work, social activity, substance use, and daily life rhythms (e.g., eating, sleeping). The variable course schedules and cyclical demands of the college academic environment heighten demand for organizational skills and long-term planning. Simultaneously, the social environment of college brings increased availability of immediate, short-term rewards. Thus, executive demands increase just as contextual support for effective executive control decreases.

"Double-Deficit" in Self-Regulation

For college students with ADHD, these increased demands on executive control occur at a time of "double deficit" in self-regulation. First, as described earlier, individuals with ADHD tend to have poorer executive functioning and a stronger aversion to delay in reward than those without ADHD. Second, their executive functioning and motivational systems (e.g., delay of reward) are not yet fully mature and are particularly susceptible to the effect of high socio-emotional arousal. The cumulative effect on immediate self-regulation task performance stemming from these two independent self-regulation deficits has not been directly tested and requires further study. However, evidence suggests that executive functioning capacity and task demands interact such that more demanding tasks (e.g., response inhibition in the presence of peers) results in greater performance decrement for those with weaker executive capacity (Gardner and Steinberg 2005). Thus, it is possible that this "double-deficit" in self-regulation increases the likelihood that the self-regulation capacity of college students with ADHD will be overwhelmed when faced with the increased executive demands and contextual challenges of college.

Fedele et al. (2010) conducted a factor analysis examining predictors of functional impairment among emerging adults and adults with ADHD. In addition to DSM-IV criteria, a broader range of executive function items uniquely predicted dysfunction among emerging adults. These results support the assertion that executive impairment is particularly problematic among college students with ADHD, whose executive processes are less well developed than their adult counterparts.

Therefore, when compared to interventions for ADHD in adults, effective intervention for ADHD among college students must lean more heavily on moderating the self-regulation demands of the environment. This can be accomplished both by increasing external support structures and decreasing challenging contextual factors. Strategies may include activating a network of support people in the immediate environment, using public commitments to support follow-through, reducing interpersonal distractions in the immediate environment, and making plans and decisions in a context of low socio-emotional arousal.

Summary of Etiology, Treatment Targets, and Contextual Issues for ADHD in College

ADHD behavior is strongly rooted in key neurological systems, particularly those underpinning response inhibition and working memory (e.g., the frontostriatal pathway) and motivation toward delayed reward (e.g., the mesolimbic pathway). Functional neuroimaging research has consistently shown underactivity in both of these networks among individuals with ADHD. Behavioral and structural neuroimaging studies further support the conclusion that these neural systems are implicated in ADHD. Furthermore, research has consistently shown that the etiology of ADHD is distinctly heterogeneous. In each individual, ADHD criterion behaviors may result from impairment in any combination of several neurological systems, and from subsequent developmental transactions between the individual and the environment over time.

Multiple Pathways to ADHD

"Cool" Executive Functions—Response Inhibition and Working Memory

Deficits in response inhibition can lead to impaired ability to sustain attention (e.g., distractibility), regulate behavioral impulses (e.g., impulsivity and emotion dysregulation), and disengage from ongoing behavior (e.g., hyperfocus). These difficulties often lead to impairment in higher-order executive processes such as organization and planning. Working memory deficits may include impairment in verbal/auditory working memory and non-verbal/visuospatial working memory. Although individuals with ADHD tend to show deficits in both types of working memory, impaired academic functioning among college students with ADHD is most strongly associated with deficits in auditory working memory. Over time, individuals with deficits in response inhibition and/or working memory may experience more task failure and punitive responses to performance on tasks requiring executive functioning. This may lead to an aversive association with and avoidance of executive tasks and to diminished opportunities to practice executive-related skills. As a result, by the time they reach college, students with ADHD may experience functional impairment on a broad range of executive tasks.

"Hot" Executive Functions—Delay Aversion

Among many individuals, ADHD-related behavior may stem primarily from impaired signaling of long-term reward, resulting in an atypical shift in motivation toward smaller, immediate rewards rather than larger, delayed rewards. This may manifest as a powerful aversion to "delay-rich" environments (e.g., low boredom tolerance) and behavioral avoidance of the experience of boredom either by avoiding boredom-provoking scenarios entirely or by engaging in behavior to generate more immediate reward (e.g., daydreaming, fidgeting). Over time, delay aversion can lead to generalized avoidance of delayed reward situations and diminished opportunities to develop skills for effectively organizing and motivating behavior toward long-term goals.

Temporal Processing

ADHD criterion behavior may also result from atypical perception of time intervals. This impairment may result in poor awareness of the passage of time or in inaccurate estimation of time required to accomplish a given task. The dramatic increase in flexibility of daily schedule from high school to college environments may prove especially problematic for individuals who struggle to effectively manage time.

Cognitive Energy State

For some individuals with ADHD, impairment in executive functioning may be related to deficits in mobilizing cognitive energy (e.g., effort, arousal) to support these neural systems. This may lead to impaired or inconsistent performance on tasks requiring input from these systems. The college environment poses many unique challenges to maintaining an optimal physiological state to support efficient cognitive functioning.

Intervention Targets

Among college students with ADHD, impairment in the neural systems described above gives rise to adaptive impairment in several areas, including academic performance, social interactions, and mood regulation.

Academic Performance

College students with ADHD tend to have lower GPAs and higher rates of academic probation and dropout. Research has consistently found that inattentive/disorganized behaviors, but not hyperactive/impulsive behaviors, are associated with impaired academic performance.

Social Interactions

College students with ADHD tend to report more difficulty sustaining attention during conversations, inhibiting impulsive interruptions or hurtful comments, and expressing anger effectively. Students with ADHD report lower social adjustment and greater social dissatisfaction. They may have more difficulty in both friendships and dating relationships and receive lower ratings from confederates after brief interactions.

Psychological Health

College students with ADHD are at higher risk for depressive symptoms, substance abuse, and overall psychological distress.

Persistence into Adulthood

Impairment in each of these domains tends to persist across the lifespan among individuals with ADHD, highlighting the importance of supporting adaptive development during emerging adulthood.

Developmental and Contextual Issues

The intersection of ADHD, emerging adulthood, and the college environment yield a unique set of challenges that must be addressed in order to effectively treat ADHD among college students. College students with ADHD face a "double-deficit" in self-regulation (e.g., executive functioning, planning) due to the conjunction of ADHD-related impairment and still-maturing neural networks supporting response inhibition and long-term planning. At the same time, for most students, the transition to college brings an abrupt decrease in external support and a marked increase in powerfully distracting environmental stimuli. In short, the college context places high demands precisely on those areas where emerging adults with ADHD tend to be most vulnerable. This unique confluence of developmental and contextual factors distinguishes college students from both adolescents and adults with ADHD, and from emerging adults with ADHD who do not attend college.

Empirical Support for Treatment of ADHD in Adults

College students with ADHD experience a unique combination of developmental and contextual challenges, suggesting that an intervention uniquely tailored for this population will be most effective. Given the lack of treatment research on ADHD among college students, we propose that treatment development work be guided by empirically supported treatment models and principles from proven interventions for adults and adolescents with ADHD. Typical-aged college students (18–24) function similarly to adults in many ways; most importantly, their day-to-day behavior is largely independent of external control, including few proximal restrictions on academic behavior, daily life routines, and substance use. On the other hand, typical-aged college students share key similarities with adolescents, including a still-maturing executive function system and a heightened vulnerability to the effects of socio-emotional arousal on decision-making.

Thus, important elements must be drawn from treatment research with adults and with adolescents.

Stimulant medication is an effective first-line treatment for many adults with ADHD; however, stimulants are intolerable or ineffective in up to 50 % of adults with ADHD, and most adults who do respond to medication still experience significant deficits in executive control and other areas of functioning (Wilens et al. 2002). Psychosocial interventions for adults with ADHD have shown promising results in initial efficacy trials. These interventions include individual CBT (e.g., Safren et al. 2010b), group-based CBT (e.g., Solanto et al. 2010), group-based mindfulness training (Zylowska et al. 2008), and individual coaching (e.g., Allsopp et al. 2005). In general, results suggest that psychosocial interventions are feasible, acceptable, and decrease core symptoms and/or functional impairment among adults who use stimulant medication and those who do not. Limited treatment research for adolescents with ADHD supports the use of behavioral family-based and school-based interventions, rather than individual CBT. These findings suggest that leveraging contextual support will likely be an important component of effective intervention with college students.

Pyschopharmacological Interventions

Stimulant medication is the most common first-line treatment for ADHD among individuals of all ages. In randomized trials, stimulants have proven effective in reducing core symptoms of ADHD among young adults and adults (Wilens et al. 1998) and improving health-related quality of life (Weiss et al. 2010). However, 20–50 % of adults in these trials were considered non-responders due to intolerable side effects or insufficient change in core symptoms. Additionally, most adults who respond to medication experience less than a 50 % reduction in symptoms, and key areas of functional impairment remain (Wilens et al. 2002). Among adults already receiving stimulant medication for ADHD, individual CBT has been shown to significantly decrease residual symptoms and functional impairment (Safren et al. 2005, 2010b). Limited research with non-stimulant medication (e.g., atomoxetine, bupropion) has shown generally lower benefits and higher side effects than stimulant medication (Wigal 2009).

No controlled studies have assessed the effectiveness of psychopharmacological interventions among college students with ADHD. One retrospective chart-review investigation found positive effects for stimulant medication treatment among college students with ADHD (Heiligenstein et al. 1996). Conversely, a large correlational study found no overall association between current stimulant medication use and academic performance among college students (Rabiner et al. 2008). These data are consistent with the adult ADHD literature that demonstrates that individuals with ADHD lack key self-regulation skills (e.g., organization, planning) that cannot be remediated by medication alone. In addition, many students report significant costs associated with stimulant use (e.g., “dull,” “zoned out,” less sociable, emotional lability, reduced appetite, difficulty sleeping) and often choose not to take medication every day as a result (Meaux et al. 2006). However, methodological limitations of the available data preclude any conclusions about the effectiveness and acceptability of stimulant medication among college students with ADHD. Randomized controlled trials of psychopharmacological interventions with this population are urgently needed.

High risk of stimulant misuse or diversion is also an important concern. Rates of illicit use of stimulant medication among college students appear to fall between 5 and 15 % (e.g., Hall et al. 2005; McCabe et al. 2005; Weyandt and DuPaul 2008), but estimates range as high as 35 % at a one small, competitive college (Low and Gendaszek 2002). Up to 70 % of students report knowing peers who misuse prescription stimulant medication (Carroll et al. 2006). The primary source of illicit stimulant medication appears to be fellow students who have been prescribed the medication for ADHD (McCabe and Boyd 2005), with up to 29 % of students prescribed stimulants for ADHD reported that they have given or sold their medication to peers (Upadhyaya et al. 2005).

Thus, although stimulant medication is a key component of treatment for ADHD, significant concerns remain regarding its overall cost-benefit profile. These concerns underscore the need for effective psychosocial interventions for college students with ADHD.

Individual CBT

A handful of studies over the past 5 years have evaluated the efficacy of individual CBT for adults with ADHD. Most follow a similar treatment model of 10–16 weekly 1-h sessions including components such as (1) psychoeducation, (2) enhancing motivation for change, (3) training and practice of skills/coping strategies, (4) cognitive restructuring of maladaptive thoughts, and (5) accessing/utilizing support. Results suggest that this approach is feasible, acceptable to clients, and efficacious in reducing core symptoms of ADHD as well as symptoms of anxiety and depression. Pre- to post-treatment effect sizes appear to be large or very large and drop to the moderate range when compared to an active control condition. Treatment gains appear stable across a 1-year follow-up period.

Safren et al. (2005) randomized 31 adults (age $M = 45.5$; range 23–59) with ADHD who were stable on stimulant medication into two treatment conditions: (1)

CBT plus continued medication or (2) continued medication alone. Participants who received CBT showed lower independent-evaluator-rated symptoms of ADHD, depression, and anxiety, with very large effect sizes ($d = 1.2$–1.4). In a subsequent large-scale follow-up trial (Safren et al. 2010b), 86 adults (age $M = 42.3$) with ADHD who were stable on stimulant medication were randomized to receive either: (1) CBT plus continued medication or (2) relaxation and educational support plus continued medication. The CBT group showed roughly twice as many responders as in the relaxation group (67 vs. 33 %), as well as greater average improvement by the end of treatment ($d =$ 0.53–0.60). Treatment gains were maintained at 6-month and 12-month follow-up assessments.

Similarly positive results have been reported from an open trial of individual CBT plus medication (Rostain and Ramsay 2006), and from a randomized trial comparing individual CBT to computer-based cognitive training and waitlist control conditions (Virta et al. 2010).

Group CBT

Several studies over the past decade have examined the efficacy of group CBT for adults with ADHD. Many of the components of these group interventions parallel those of the individual CBT programs described above; one intervention included mindfulness training as well as the more standard components (Philipsen et al. 2007). The group format permits less individually targeted intervention, but provides added benefits of peer normalization, support, and modeling, as well as greater cost-effectiveness. Results suggest that group CBT is acceptable, feasible, and efficacious in reducing both primary and secondary symptoms of ADHD. As with individual CBT, pre- to post-treatment effect sizes appear to be large or very large and remain in the moderate range when compared to an active control condition. Treatment gains appear to be maintained across a 1-year follow-up period.

Solanto et al. (2008) treated 30 adults with ADHD (age $M = 41.8$; range 23–65) with a group CBT approach in an open trial. The program consisted of either 8 or 12 weekly 2-hour sessions. Pre- to post-treatment effect sizes were moderate on core ADHD symptoms ($d = 0.59$–0.67). In a subsequent randomized trial (Solanto et al. 2010), 88 adults with ADHD (age $M = 41.0$) were assigned to receive either: (1) group CBT or (2) supportive group therapy. Both programs consisted of 12 weekly 2-h sessions. The supportive therapy intervention included psychoeducation, validation, and group discussion of specific ADHD-related topics, but no didactic strategies or exercises. Compared to the supportive control condition, group CBT showed greater improvement in core ADHD symptoms ($d = 0.67$) and a higher response rate (53 vs. 28 %). Medication status was not associated with response to treatment. Treatment response was moderated by baseline symptom severity: Change scores among subjects receiving supportive therapy were consistent across all levels of baseline severity, while higher baseline severity was associated with greater improvement among subjects receiving group CBT.

Philipsen et al. (2007) conducted a multi-site open trial in which 72 adults (age $M = 35.8$; range $=$ 18–53) received group CBT. Hesslinger et al. (2002) had previously reported promising results from a small open pilot trial ($n = 8$) using this intervention. The treatment consisted of 13 2-h sessions and combined elements of dialectical behavior therapy (DBT; Linehan 1993) and traditional CBT in a skills-training format. Pre- to post-treatment effect sizes were large on the primary (partial $\eta^2 = 0.411$) and secondary outcomes (partial $\eta^2 = 0.314$). Session topics of behavior analysis, mindfulness, and emotion regulation were rated by subjects as the most helpful. Medication status and treatment site did not predict outcomes.

In addition, several other evaluations of group CBT have shown evidence of efficacy. In a randomized trial, Stevenson et al. (2002) found higher core symptom response rates in a CBT group (55 %) than a waitlist control group (36 %). Notably, this intervention included the unique component of a "support person" who promoted subjects' attendance and engagement in treatment. In an open trial, Virta et al. (2008) found moderate-to-large pre- to post-intervention improvements, with treatment gains maintained throughout a 6-month follow-up (Salakari et al. 2010). Wiggins et al. (1999) also showed pre- to-post-treatment improvements in an open trial.

Mindfulness Training

Research conducted over the past 5 years suggests that mindfulness training may be an efficacious intervention for ADHD in adolescents and adults. Given that inattentive symptoms appear to be the primary driver of functional impairment among both college students and adults with ADHD, the regular practice of purposeful, non-judgmental attention to the present moment may be a valuable component of effective intervention with these populations.

Adults with ADHD have lower self-reported mindfulness skills than normal controls (Smalley et al. 2009). In particular, subjects with ADHD report greatest relative difficulty ($d = 1.35$) on the "acting with awareness" dimension—attending to current actions rather than behaving automatically or absent-mindedly (e.g., "I don't pay attention to what I'm doing because I'm daydreaming, worrying, or otherwise distracted"). Adults with ADHD also reported lower scores on the "describing" ($d = 0.49$) and "nonjudging of inner experience" ($d = 0.62$) dimensions. Carmody and Baer (2008) found that time spent in mindfulness practice predicted pre-post changes in the "observing," "acting with awareness," "nonjudging of inner experience," and "nonreactivity to inner experience" among adults who participated in a mindfulness-based stress reduction program (MBSR; Kabat-Zinn 1982), with moderate-to-large pre- to post-treatment effect sizes ($d = 0.47$–0.91). Notably, time spent doing both sitting meditation and movement meditation (e.g., yoga) was associated with improvement in acting with awareness.

In a well-controlled, randomized study, Tang et al. (2007) assigned 40 Chinese undergraduates to receive either an integrative sitting and movement meditation practice or relaxation training. Following training in the practices, subjects practiced in a group setting for 5 days, 20 min/day. After only 1 week of practice, students in the meditation training group outperformed relaxation training subjects on the Attention Network Test (ANT; Fan et al. 2002), a computerized test designed to assess conflict attention. The ANT has been previously shown to activate key components of the frontostriatal circuit (particularly the lateral prefrontal and anterior cingulate cortices) that support executive functioning. The meditation training group also showed increases in positive mood and immunoreactivity and decreases in anxiety, depression, anger, fatigue and levels of stress-related cortisol.

To our knowledge, three published trials have included mindfulness training either as a primary intervention or intervention component for adults with ADHD. Zylowska and colleagues (2008) conducted an open trial of a mindfulness training program for 24 adults (age $M = 48.5$) and 8 adolescents (age $M = 15.6$) with ADHD. The program was adapted from MBSR (Kabat-Zinn 1982), consisting of eight weekly 2½-h group sessions. Sessions included mindfulness practice, homework review, and group discussion, as well as psychoeducation about ADHD etiology and reframing of ADHD as a neurobiological difference that carries both maladaptive and adaptive aspects. Thirty percent of subjects were classified as treatment responders (at least 30 % reduction in ADHD symptoms). Significant improvement was found in conflict attention and set shifting, as well as symptoms of anxiety and depression. The intervention received high satisfaction ratings from adolescents as well as adults.

One group CBT intervention for adults with ADHD (Hesslinger et al. 2002; Philipsen et al. 2007) included mindfulness training and daily mindfulness practice as a key component of the treatment. (See *Group CBT* section for additional details.) Improvements resulting from the program cannot be attributed specifically to the mindfulness component; however, the data suggest that mindfulness training is acceptable and feasible as an element of a broader skills-training approach for adults with ADHD. Mindfulness ranked second on the list of most helpful components of the program based on post-treatment assessments.

An investigation of movement-based mindfulness training (e.g., pilates, tai chi) among college students found that mindfulness skills improved as a result of this training, and that subjects experienced improved mood, perceived stress, and sleep quality (Caldwell et al. 2010).

Thus, the combination of basic and applied research on mindfulness skills and ADHD suggests the following: (1) adults with ADHD show significant deficits in mindfulness skills, particularly in "acting with awareness"; (2) mindfulness practice strengthens these skills and improves performance on neuropsychological tests of attention performance; (3) interventions including mindfulness training (either as a stand-alone or as part of a larger intervention) lead to reductions in inattentive symptoms among adolescents and adults with ADHD; and (4) mindfulness training leads to collateral benefit in areas often affected by ADHD (e.g., mood regulation).

Coaching

ADHD coaching, although encompassing a fairly broad array of supportive services, is typically described as a focused relationship designed to help clients implement effective strategies to manage their daily lives. Although ADHD coaching has gained significant popular attention over the past 15 years, including numerous ADHD coach "certification" programs, very little research evaluating its efficacy has been published (Goldstein 2005). To date, research has not clearly differentiated the components of ADHD coaching from individual CBT treatment.

Allsopp et al. (2005) examined the efficacy of study strategies coaching in an open trial with a sample of 46 college students with ADHD. Subjects received 2–6 h of individual coaching, including general guidance in organizational skills (e.g., daily planner use) and tailored instruction in skill areas of need (e.g., reading comprehension). Moderate effect size changes in GPA were observed from pre- to post-treatment; these effects were maintained into the next semester, although some students continued to receive coaching throughout follow-up. Zwart and Kallemeyn (2001) reported promising results from a peer-based coaching program for college students with ADHD and/or learning disabilities, suggesting that this approach may be useful in supporting adaptive coping strategies and study skills among college students. Kubik (2010) reported pre- to post-intervention improvement among a group of 45 adults with ADHD who attended a large group workshop. The workshop included six weekly 2-h sessions and a follow-up 2-h session 1 month later.

Although described as a coaching intervention, this program appears to be more similar to group CBT skills approaches described above than to typical one-on-one ADHD coaching interventions.

Psychosocial Interventions for Adolescents with ADHD

Psychosocial treatments showing preliminary evidence of efficacy for adolescents with ADHD fall into two general categories: family-based interventions and school-based interventions (see Chronis et al. 2006, for a detailed review). A cognitive-oriented approach showed no efficacy in one clinic-based trial (Morris 1993).

In family settings, behavior management training, problem-solving and communication training, and structural family therapy programs have all shown pre- to post-treatment improvement in several key areas (Barkley et al. 1992, 2001), as did a group-based parent training program (McCleary and Ridley 1999); however, conclusions based on these findings are tempered by lack of a control group and failure to demonstrate clinically significant reliable change at follow-up.

An intensive school-based intervention outperformed standard community care in a randomized trial (Evans et al. 2004, 2005). The intervention focused on behavioral strategies for teachers, counseling, and parents to implement, including a daily report card, time out, contracting, a point system, and programs to enhance shaping and generalization of effective skills. In addition, adolescents received instruction in study skills (notetaking, organization, problem solving) and basic social skills as well. Relative to the control group, this program showed moderate-to-large effect sizes in improvement on measures of academic and classroom functioning. In addition, single-subject studies have suggested that classroom-based functional analysis and self-monitoring may improve on- and off-task behavior (e.g., Ervin et al. 1998).

Although only a small number of studies are available, research on treatment for ADHD among adolescents suggests that external management of behavioral contingencies is a key component. This conclusion further emphasizes the challenge presented by the loss of family- and teacher-provided contingency management as students with ADHD transition to college. Although intervention with college students will look quite different from intervention with adolescents, leveraging contextual supports must remain an important theme. College students with ADHD may need to learn how to generate external contingencies for themselves (e.g., a study group), rather than relying on parents or teachers to create these contingencies for them.

Proposed Treatment Targets and Principles for Intervention with College ADHD

As noted above, ADHD among college students is associated with functional impairment across several domains and greater risk for negative outcomes such as academic probation and school dropout. Unfortunately, to date, there have been very few published studies on psychosocial treatments for ADHD among college students. There is an urgent need for treatment development research in this area. As a guide both for ongoing treatment efforts and for treatment development efforts, empirically based intervention targets and principles are needed. In service of this goal, the following section combines three lines of research: (1) basic research on impairment among college students with ADHD, (2) etiology and developmental context for ADHD in emerging adulthood, and (3) intervention research for adults with ADHD. In the following section, proposed intervention targets are discussed, along with corresponding treatment principles/approaches for each target.

It is very important to note that although the following recommendations for intervention are grounded in best available research, they are virtually untested in the college population. The reader is encouraged to view these as tentative hypotheses in need of careful study.

Target: Inattentive/Disorganized Behavior

Inattentive/disorganized symptoms have been shown to account for significant variance in overall adaptive functioning, academic concerns, depressive symptoms, and GPA; conversely, hyperactive/impulsive symptoms do not appear to predict differences in adaptive functioning. Inattentive/disorganized symptoms hinder academic functioning in several ways: (1) difficulty sustaining attention during class, reading, and other activities, leading to impaired learning of necessary knowledge and skills; (2) difficulty organizing, initiating and sustaining motivation and effort on academic tasks, leading to late, incomplete, or lower-quality work; and (3) poor time management, including difficulty managing sleeping and waking times, leading to lateness or absences from class, meetings, or other activities (e.g., Meaux et al. 2009). Impairment in several neurological domains may give rise to inattentive/disorganized behavior and each may require tailored intervention.

Addressing Executive Function Deficits (Response Inhibition)

Individuals with ADHD may have difficulty along several subdomains of response inhibition: (1) inhibiting a prepotent response in order to act with awareness (e.g., ignoring a text message alert on a cell phone while studying); (2) interrupting an ongoing response in favor of a more adaptive response (e.g., stopping web surfing and returning to studying); and (3) interference control (e.g., blocking extraneous stimuli such as conversations or other thoughts in order to continue studying). These basic processes serve to modulate and direct attention on a moment-to-moment basis and also underlie higher-level executive functions such as planning, organization, and prioritization.

Among the published interventions for ADHD in adulthood, nearly all include strategies for addressing deficits in response inhibition and related "cool" executive functions. These strategies tend to fall into two general categories: (1) improving executive control and (2) reducing environmental demands on executive functioning systems.

Strategies for improving executive control include:

Mindfulness Practice Mindfulness practice has been shown to increase attention performance after as little as 1 week. Sustained practice over several weeks improves acting with awareness and interference control, thereby decreasing automatic behavior and distraction. The dose–response relationship of mindfulness practice and attention control has not yet been clarified. However, results from a mindfulness-based treatment program suggest that this approach is acceptable to both teens and adults with ADHD, and that 10–20 min of daily practice is realistic and efficacious (Zylowska et al. 2008).

Externalizing Higher-Level Executive Processes The majority of efficacious interventions for adult ADHD include skills/strategies for developing external supports for higher-level executive processes such as organization, prioritization, and planning. Key strategies include: (1) using a calendar/planner and notepad to organize all important information needed for daily life functioning; (2) breaking tasks down into small, manageable chunks and using a notepad list to prioritize tasks; (3) using a notepad to record distractions during work time rather than immediately acting upon them; and (4) using timers/alarms to support effective use of time. Most efficacious programs for adults with ADHD devote multiple treatment sessions to these skills so that new routines can be effectively shaped and rehearsed.

Strategies for reducing environmental demands on executive functioning include:

Reducing Distracting Stimuli Engineering the environment to minimize competing stimuli can help to reduce the load placed on the response inhibition system. Strategies may include: (1) studying in a location with few external distractions; (2) limiting access to electronic distractions such as cell phone and internet; and 3) writing down important but distracting thoughts to reduce the need to hold them in mind.

Amplifying Relevant Stimuli Similarly, adapting the environment to make relevant stimuli more "sticky" can reduce response inhibition load. Strategies may include: (1) placing the calendar/notepad in an obvious place; (2) placing objects to serve as their own reminders; and (3) using small notes/reminders to cue specific behavior (e.g., a small colored dot on a computer monitor to cue effective use of computer time).

Addressing Executive Function Deficits (Working Memory)

Deficits in both auditory/verbal working memory (i.e., phonological loop) and visual-spatial working memory (i.e., visuo-spatial sketchpad) are common among college students with ADHD. Although visual-spatial working memory has shown more robust deficits among children with ADHD, preliminary research with college students suggests that weakness in auditory/verbal working memory in particular predicts lower GPAs (Gropper and Tannock 2009). This may be due to the predominantly verbal presentation of much of the material in college courses (e.g., lecture courses and reading). Students with deficits in working memory may have greater difficulty integrating multiple ideas and processing information on a deeper level that facilitates long-term recall. They may also frequently forget important details in daily life because that information was "bumped" out of working memory before it could be utilized (i.e., failing to recall an assignment, a meeting, or a new acquaintance's name because it was never encoded into long-term memory).

Unlike response inhibition, working memory itself does not appear to be improved by mindfulness practice (e.g., digit span does not increase); however, many of the executive function support strategies facilitate more efficient use of available working memory. Possible strategies to emphasize for students with working memory deficits include: (1) *immediately* writing down all important information in planner/notepad; (2) using a notepad to briefly list distracting thoughts for later attention; (3) using externalized representations of information (e.g., mind maps) to facilitate integration of information and ideas; and (4) repeating names of new acquaintances in conversation.

Addressing Executive Function Deficits (Delay Aversion)

Due to impaired neurological signaling of delayed rewards, individuals with ADHD may have particular difficulty in motivating behavior toward larger, long-term rewards over smaller, short-term rewards. This aversion to delayed rewards may manifest in several different ways: (1) intentionally choosing (rather than automatically, as with response inhibition failure) a smaller, immediate reward (e.g., talking with friends, playing computer games) over a larger, long-term reward (e.g., starting a paper that is due in a week, going to sleep in order to be well rested for class the next day); (2) low boredom/frustration tolerance and general avoidance of situations that require sustained effort for reward; and (3) under-developed organization and planning skills due to insufficient or punitive prior experience with managing delay and delay-demanding activities. All of these may lead to chronic procrastination and resultant poor performance.

Strategies for addressing deficits in motivation toward long-term rewards may include:

Structuring Contextual Factors The effect of socio-emotional arousal on decision-making processes has important implication for addressing delay aversion among college students with ADHD. As described by Steinberg (2007), emerging adults (age 18–24) perform as well as adults (age 25+) in weighing short- and long-term consequences of behavior while in a state of low socio-emotional arousal (i.e., calm and alone); however, emerging adults show more risky behavior and greater preference for immediate rewards while in the presence of peers. Most college students live in environments with high exposure to peers. As a result, interventions that target self-regulation (e.g., contingent self-reinforcement) may be less effective among college students than among adults due to difficulty with follow-through. Conversely, interventions that effectively leverage the power of socio-emotional context may prove even more effective among college students than adults.

Establishing External Reinforcement Contingencies Because effective self-reinforcement is challenging, it may be more effective to create external structures to reinforce effort toward long-term rewards. Examples of this include public commitments to others who will hold the individual accountable (e.g., pledging to roommates to be in bed by midnight on weeknights), scheduling with others to engage in an activity together (e.g., studying with a friend), or joining an organization that supports a desired activity (e.g., signing up for a weekly yoga class or joining an intramural sports team). Students may also find it helpful to enlist help from a specific “support person” (e.g., friend, tutor) to provide regular external accountability. Discussing time management challenges with instructors early in the term may allow students to structure periodic check-ins, or even smaller, more frequent deadlines.

Breaking Tasks into Small Chunks The delay in initiation of effort toward long-term reward increases with the length or amount of effort required to reach that reward (e.g., a student will tend to delay starting a 10-page paper longer than a 2-page paper). One component of most executive function strategies is the skill of breaking tasks into chunks that can be accomplished quickly, resulting in more rapid and more frequent natural reinforcement from task completion.

Contingent Self-Reinforcement Immediate self-reinforcement for completing each task can support sustained effort toward a larger, delayed reward (e.g., reinforcing every 10 min of work with 2 min of break time).

Pros and Cons of Working Toward Long-Term Reward Although often used as a decision-making strategy, the process of writing a list of pros and cons may support motivation as well. Key elements of this approach are to identify 1–3 reinforcers that are most compelling in motivating behavior toward the desired goal and to implement strategies to make distal rewards more proximal (e.g., visualizing the long-term reward or the aversive consequences of failing to achieve the long-term reward, creating visual reminders of the long-term reward).

Mindfulness Practice Regular practice of a boredom-eliciting activity (e.g., sitting meditation) without avoiding or escaping may partially extinguish or attenuate the aversive reaction to the physical sensations associated with boredom. This, in turn, may reduce behavioral avoidance of daily life activities that elicit boredom due to delayed reward.

Addressing Temporal Processing Deficits

Individuals with ADHD tend to experience atypical perception of time, including less accurate estimation of time and reproduction of time intervals. In some cases, these difficulties may overlap or interact with executive dysfunction or greater delay aversion; however, many individuals who meet criteria for ADHD show impaired temporal processing but typical response inhibition, working memory, and delay aversion.

Strategies for addressing deficits in temporal processing may include:

Time Reminders Using timers and alarms to provide external markers of time may be an effective coping strategy (e.g., using a timer to limit study breaks; setting a “bedtime alarm” to cue the end of evening activities).

Planning Backward In order to effectively plan important sequences of behavior that must be carried out in a timely fashion, beginning with the end result and planning in reverse temporal order, including buffer time for unexpected problems, may be effective (e.g., to be reliably on time for a 9:30 a.m. class, plan to arrive at the room by 9:20 a.m., which means walking out the door at 9:05 a.m., which requires eating breakfast by 8:45 a.m.). Timers or alarms can be used in conjunction with this approach to ensure that the plan is followed.

Summary

Inattentive/disorganized behaviors appear to be the primary driver of functional impairment among college students with ADHD. There is significant heterogeneity, however, in which specific inattentive behaviors are most problematic, and the associated neurobiological underpinnings. When assessing ADHD symptoms, clinicians should bear in mind the key domains of response inhibition, working memory, delay aversion, and temporal processing, and attempt to gauge the degree to which the proximal behaviors map onto these domains. Clinicians may find it useful to discuss these possible pathways when delivering psychoeducation on the etiology of ADHD. By applying this conceptual framework to the process of treatment, it may be possible to more efficiently identify key problematic behaviors and more effectively target interventions to those behaviors.

Target: College Academic-Specific Skills

For most students, with or without ADHD, effective functioning in college demands the acquisition and practice of a new set of academic skills specific to the college environment. Students with ADHD tend to have less effective study skills in key areas than both typical students and students with learning disabilities. These key areas of deficit are: (1) time management, (2) concentration, (3) selection of main ideas, and (4) test-taking strategies.

Time Management

Among college students in general, time management may account for more variance in GPA than do SAT scores (21 vs. 4 %; Britton and Tesser 1991). In particular, students who engage in more structured short-term planning (e.g., "Do you make a list of things you have to do each day?") and who follow through on study plans (e.g., "Do you make constructive use of your time?") tend to show higher GPAs. Notably, time management appears to account for less than 10 % of variance in GPA among high school students (Xu 2010). The day-to-day workload of many college courses varies widely over the course of the academic term and may contribute to the heightened demand for effective time management in college.

Given that college students with ADHD tend to have both poorer time management skills (Reaser et al. 2007) and lower GPAs, and that time management predicts GPA among college students in general, teaching students with ADHD to effectively manage time in the college context may be beneficial. This may include strategies for scheduling work (e.g., reviewing each syllabus and mapping out all assigned work for the term) and for structuring contingencies to support follow-through (e.g., working with a study group). Further research is also needed to directly examine the potential mediating role of time management behavior in the relation between ADHD and GPA among college students.

Concentration

Strategies for improving concentration are discussed in detail above, including mindfulness practice and structuring the environment effectively.

Selection of Main Ideas and Test-Taking Strategies

Difficulties with response inhibition, working memory, and delay aversion may all contribute to difficulty in identifying main ideas and using effective test-taking strategies. Many students with ADHD struggle with these skills in high school but manage to compensate well enough to succeed; however, the increased demands of college course material and exams may be overwhelming without effective study skills in these areas. Effective intervention for ADHD in college should include assessment and training in effective use of these study skills. For example, the RAP paraphrasing strategy (R—Read a paragraph; A—Ask what the main ideas are; and P—Put the ideas into your own words) (Schumaker et al. 1984) was used in an educational intervention showing efficacy among college students with ADHD (Allsopp et al. 2005).

Target: Disrupted Daily Life Functions

For many students, the college environment increases risk for disrupted daily life routines in several domains, particularly sleep, eating, exercise, medication adherence, and substance use. Many external contingencies that govern these behaviors among high school students (e.g., living with parents/family, greater structure in daily school schedule, participation in high school sports) are no longer present after the transition to college. Although dysregulated daily life patterns are detrimental to academic performance for all students, individuals with ADHD are at heightened risk for adverse outcomes (e.g., academic probation, dropout) due to ADHD-related deficits in executive functioning and other areas. Relative to adults with ADHD, college students with ADHD are more prone to this type of dysregulation in daily life routines due to their unique developmental and psychosocial context. Supporting healthy daily life behavior is a key target of effective intervention with this population.

Balanced Sleep

In general, college students tend to have later sleep schedules, more irregular sleep patterns, and get less sleep than recommended (Buboltz et al. 2006). College students with ADHD are at higher risk for unhealthy sleep patterns in several ways: (1) young adults with ADHD are two to five times more likely to experience a wide range of sleep disturbances, including difficulty falling asleep, restless sleep, periodic leg movements, and sleep-disordered breathing (Gau et al. 2007); (2) a common side effect of stimulant medication use is increased difficulty falling asleep; and (3) students with ADHD show higher rates of substance use and abuse, which is associated with poorer sleep patterns (particularly delayed sleep schedule) and sleep quality (Singleton and Wolfson 2009).

Unhealthy sleep patterns are associated with decreased alertness, working memory and response inhibition (including sustained attention), and mood dysregulation, and are strongly linked to poor academic performance (e.g., Durmer and Dinges 2005; Fafrowicz et al. 2010; Van Dongen et al. 2003). Sleep-related decrements in executive function and mood regulation may be especially detrimental to individuals with ADHD who already struggle in these areas. Therefore, regulating sleep patterns should be a key target for intervention among college students with ADHD. Important strategies include: (1) psychoeducation on sleep hygiene (e.g., limit caffeine intake after noon, dim lighting and do not use electronics for 30–60 min before bedtime, do not engage in wakeful activities in bed); (2) use an alarm to cue bedtime routine; (3) moving sleep schedule earlier; and (4) reducing disruption of sleep pattern and quality due to alcohol and other substance use.

Balanced Eating

Although dietary concerns do not appear to be a primary cause of ADHD symptoms in the vast majority of cases, research suggests that a subset of individuals may benefit from dietary changes. Studies using an oligoantigenic diet (“few foods” or “elimination diet”) with children have shown relatively high response rates (e.g., Egger et al. 1985), including up to 64 % of intent-to-treat subjects (or 78 % of program completers) in a recent well-controlled study (Pelsser et al. 2011). However, no controlled studies have been published exploring dietary interventions for adults with ADHD. This is an important area for further research. For the present time, young adults with ADHD who do not respond to primary treatments or who desire a dietary intervention may be advised to do so under proper guidance.

Additionally, emerging research suggests that adequate blood glucose levels are required for optimal function of self-control across a wide variety of tasks (e.g., attention control, emotion regulation). Exertion of self-control depletes blood glucose, diminishing capacity for the next episode of self-control; replenishment of blood glucose can remediate this deficit (Gailliot et al. 2007). Research with children has found that glucose intake leads to immediate improvements in short-term performance on certain executive functioning tasks, including those requiring sustained attention or conflict attention (e.g., Stroop) (e.g., Benton et al. 1987). Although further research is needed to explore this apparent effect, students with ADHD may find it useful to experiment with eating breakfast (if they do not already do so) or eating a small, glucose-rich snack before periods of sustained attention or effort.

Balanced Exercise

Physical activity has been shown to improve executive functioning among children with ADHD immediately after exercise, whether treated by stimulant medication or not (Medina et al. 2010). Additionally, regular exercise may lead to long-term improvement in ADHD-related behavior. In a randomized trial, children with ADHD who engaged in a 45-min exercise program 3 times weekly for 10 weeks showed improvements in parent-reported and teacher-reported attention problems as well as performance on a neuropsychological measure of sustained attention. Although the precise mechanism of these effects remains unclear, exercise increase levels of dopamine and norepinephrine—the two primary neurotransmitters targeted by ADHD medication—and brain-derived neurotrophic factor (BDNF) and insulin-like growth factor-1 (IGF-1). Both BDNF and IGF-1 enhance neurogenesis and neural plasticity, and BDNF levels have been shown to be correlated with sustained attention on continuous performance tests among adults (Halperin and Healey 2011). Research is needed to examine the effects of exercise among college students; however, it is possible that students with ADHD may find that regular exercise reduces symptoms associated with ADHD.

Medication Adherence

Adherence to prescribed stimulant medication appears to decline after the transition to college, with many college students reporting that they take medication only when work demands are high (Meaux et al. 2009, 2006). Although the effectiveness of variations in medication adherence has not been assessed among young adults or adults with ADHD, "as-needed" medication use may serve to maintain cycles of procrastination followed by intense, stimulant-supported work as deadlines approach. Promoting regular medication use among students who choose to use medication may facilitate learning and practice of organization, prioritization, and long-term planning skills. However, as discussed above, many college students report that taking medication is a "trade-off" between academic benefits and aversive side effects and may not find regular medication use to be worthwhile.

Addictive Behaviors

College students, young adults, and adults with ADHD are all at higher risk for tobacco use, alcohol abuse, and illicit drug use (Rooney et al. 2012; Upadhyaya et al. 2005). Nicotine, alcohol, and most recreational drugs stimulate dopamine activity in the central nervous system similarly to stimulant medication, and many college students report self-medicating with tobacco in part because it reduces ADHD symptoms (Meaux et al. 2009). Along with the higher risk of alcohol abuse associated with ADHD status, the developmental stage of emerging adulthood (age 18–24) and the context of college both increase risk of abuse (Substance Abuse and Mental Health Services Administration 2007). This conjunction of factors creates a "triple vulnerability" for typical-aged college students with ADHD. Alcohol use has been shown to predict poorer academic performance (Buboltz et al. 2006); notably, much of this association between alcohol consumption and cumulative GPA is mediated through dysregulation of sleep schedule (not total sleep duration) and resulting daytime sleepiness. As described above, unhealthy sleep patterns can significantly disrupt attention, executive functioning, and mood regulation.

The combined factors of greater risk of substance abuse and greater vulnerability to impaired executive function underscore the importance of addressing substance use as a component of treatment for ADHD among college students. A harm reduction approach including motivational interviewing and personalized normative feedback has been shown to be effective with college students (Carey et al. 2007) and may be a beneficial adjunct to treatment of ADHD in college.

Target: Emotion Dysregulation

Individuals with ADHD tend to show greater difficulty managing emotions than their peers without ADHD. In particular, college students with ADHD experience more depressive symptoms (Blase et al. 2009) and overall psychological distress (Richards et al. 1999) and show more ineffective expressions of anger (Ramirez et al. 1997).

Cognitive restructuring of maladaptive or depressogenic thoughts—a mainstay of CBT treatment for anxiety and depression—is a common component of many individual and group CBT interventions for adults with ADHD. In addition to improving core symptoms of ADHD, this approach may also help to reduce symptoms of anxiety and depression. Mindfulness training has been found to provide broad-based benefits in psychological functioning, in addition to its enhancement of attention control. Emotional regulation skills taught in DBT (Linehan 1993) were included as a component of a larger skills-based intervention for adults with ADHD (Philipsen et al. 2007) and were among the top three skill areas (along with mindfulness and behavior analysis) in patient-reported effectiveness.

Target: Social Relationships

College students with ADHD report lower social skills and greater difficulty in social and romantic relationships. Non-ADHD students tend to report more negative associations and unfavorable attitudes toward their ADHD peers. Impaired social functioning may result from a variety of ADHD-related behavior, including difficulty paying attention or interrupting in conversation, blurting out hurtful comments, ineffective expressions of anger or other emotions, lateness, and failure to follow through on commitments or responsibilities. Although academic and social impairment may arise independently from ADHD behavior, some have argued that difficulties with social relationships may also contribute to some college students' poor academic performance and higher dropout rates (e.g., Wolf 2001).

Training in mindfulness and emotion regulation skills can be helpful in addressing some of the ADHD-related impairment in interpersonal relationships. In addition, interventions for adults with ADHD have also taught skills for managing expectations, improving reliability, and using effective apology and repair in relationships.

College students—regardless of ADHD status—tend to endorse more negative adjectives to describe an individual with ADHD; however, greater frequency of contact with individuals with ADHD is associated with more positive attitudes toward those with ADHD. These results suggest two important implications: (1) college students with ADHD may benefit particularly from group-based treatment that facilitates peer normalization and (2) greater openness about the experience of living with ADHD may ameliorate some of the negative bias toward ADHD that exists among peers without ADHD.

Target: Utilizing External Sources of Support

Results from qualitative interviews with college students with ADHD have suggested that "utilizing sources of support" is a key factor in successful adaptation to the college environment (Meaux et al. 2009). In general, this includes a transition from relying primarily on parents during high school to relying primarily on friends, teachers and tutors, and academic/disability support services in college. Specifically, students reported that external support from friends, teachers, and tutors helped them to stay organized, complete assignments and attend classes on time, and stay motivated. Positive results from a group CBT intervention involving a "support person" (Stevenson et al. 2002) and an individual coaching intervention (Allsopp et al. 2005) bolster the recommendation to assist students in connecting to individuals in their immediate environment who can offer support, particularly for effective organization and time management.

College students with ADHD also report seeking educational accommodations such as extended test time and quiet examination rooms (Meaux et al. 2009). Notably, although widely used, educational accommodations have received no rigorous empirical testing, and thus, their efficacy is unknown (Weyandt and DuPaul 2008).

Prioritizing Treatment Targets and Structuring Intervention

As discussed above, ADHD is a heterogeneous disorder, both in terms of how it arises (multiple etiological pathways) and how it manifests among college students (wide range of problematic behaviors and contexts of impairment). A host of treatment strategies have shown efficacy with adolescents and adults and require careful study with the college population. Given this broad array of behavioral presentations and intervention options, structuring treatment in either a clinical or a research setting will likely be complex. In order to flexibly respond to varied presentations, we suggest a principle-based approach in which assessment of key proximal behavior targets (e.g., procrastination, disorganization, lateness) and key underlying areas of deficit (e.g., response inhibition, delay of reward) guides the application of appropriate interventions. At present, no research exists to guide the selection of various treatment options for a particular behavioral target, or the timing of intervention (e.g., upon entry to college, as indicated by academic difficulty/probation). For example, drawing upon the adult literature, a deficit in response inhibition resulting in distractibility and procrastination may potentially respond to stimulant medication, mindfulness training, cognitive-behavioral skills training, or some combination of these or other interventions. A systematic program of research is needed to evaluate the relative efficacy of treatment options and generate evidence-based recommendations for intervention with this population.

Key Issues for Intervention Research for ADHD in College

The unique aspects of ADHD among college students raise several important challenges for intervention developers. The developmental and psychosocial context specific to this population must be specifically addressed, as well as related issues of motivation for change and long-term maintenance of treatment gains. Additionally, the cyclical academic schedule and questions of acceptability and feasibility are key factors to consider. The following provides a discussion of these challenges along with brief recommendations for intervention developers.

Effective Intervention for Emerging Adults in the College Context

As detailed above, college students with ADHD face a unique set of challenges in developing effective self-management skills. ADHD often brings difficulty with resisting distractions, inhibiting impulsive responses, and organizing behavior toward long-term goals. Additionally, emerging adults with ADHD are more vulnerable to self-management problems than adults with ADHD because key cognitive control networks (e.g., executive functioning) do not reach full efficiency until around the end of emerging adulthood (about age 25). In particular, research has shown that impulsive and ineffective decision-making among emerging adults increases in contexts of high socio-emotional arousal (e.g., when peers are present). For most students, the transition to college brings an abrupt and dramatic *decrease* in external structure and contingencies (e.g., parental rules and consequences), and a simultaneous *increase* in exposure to contexts of high socio-emotional arousal (e.g., living with peers). The intersection of these biological, developmental, and contextual vulnerabilities presents a complex challenge for college students with ADHD, and for those working to guide them toward adaptive development.

Thus, college students with ADHD may be able to successfully articulate their behavioral goals in treatment sessions, yet consistently fail to perform the goal behaviors in their real-world context. To address this problem, clinicians and researchers may find it helpful to attend to contextual factors. Relative to intervention for adult ADHD, intervention for college ADHD may benefit from increased focus on strategies for adapting contextual factors to support effective behavior (e.g., walking to a morning class with a friend; studying with a friend or tutor; asking roommates to support a bedtime goal) rather than maladaptive behavior (e.g., consistently staying up late and missing morning classes).

Enhancing and Sustaining Motivation for Change

Emerging adults—and college students in particular—experience an increase in autonomy in several domains, including self-initiation of treatment seeking. Whereas high school students often present to treatment at their parents' urging, college students may be more likely to seek treatment of their own accord. However, for some individuals, treatment seeking may still result primarily from external contingencies such as academic probation or potential loss of loan support due to low grades. Additionally, help-seeking in general appears to be an area of ambivalence for many college students with ADHD (Meaux et al. 2009).

Given the likelihood of ambivalence regarding treatment, enhancing and sustaining motivation for change will be a key aspect of effective intervention. The literature on brief motivational interventions to support change of substance use behavior among college students (e.g., Carey et al. 2007) may provide a helpful analog for this important function. Drawing upon this model, a combination of assessment, feedback, and brief psychoeducation on ADHD—including associated impairment and efficacy of treatment—may enhance initial motivation toward treatment. Strategies to help sustain motivation throughout the treatment may include enlisting an individual support person (e.g., Stevenson et al. 2002) and visualizing long-term goals (e.g., Solanto et al. 2010). Given what is known about college students with ADHD, two additional strategies may be helpful in promoting continued motivation for treatment: (1) structuring treatment to provide relatively immediate rewards for participation in treatment and (2) leveraging social context (i.e., peer support) to support engagement.

Generalizing and Maintaining Behavior Change

A common element of all efficacious psychosocial treatments for adult ADHD is the use of outside-of-session homework practice and in-session feedback/troubleshooting. This serves the function of strengthening and generalizing the newly learned skills and strategies. Given the chronic nature of ADHD, it is important to identify ways to solidify new daily habits in order to promote sustained, long-term improvement. Daily tracking of at-home skills practice serves two key functions: (1) promoting generalization of skills to daily life and (2) providing detailed information to allow for corrective feedback in the following session. Booster sessions may also be helpful in promoting long-term change. Given the cyclical academic schedule, booster sessions may be especially useful when scheduled at the start of a subsequent academic term, thus promoting the maintenance of effective daily routines previously learned.

Cyclical Demands of the Academic Term

The structured schedule of academic terms presents a unique challenge for intervention with college students. Workload tends to follow a cyclical pattern throughout an academic term, with lesser workload near the beginning of the term and greater workload near the end (and perhaps the middle). Because of their increased difficulty with procrastination, in the absence of external deadlines, students with ADHD may particularly struggle to effectively distribute study time and effort throughout the term, rather than intensely concentrated near the end of term.

Treatment developers may find it useful to attend to these cyclical academic demands when designing and testing interventions. It may be advantageous to develop and test interventions that can be delivered within a single academic term—approximately 9–10 weeks for schools operating on the quarter system. Moreover, student reports of academic and psychological functioning may naturally fluctuate throughout the term along with the variation in workload. As such, it may be important to conduct assessments for all students at roughly the same point in the term.

Intervention Feasibility and Acceptability

Given the lack of intervention research for college students with ADHD, feasibility and acceptability are important matters to be addressed in initial treatment development efforts. Individual CBT, group CBT, and mindfulness training have all shown good acceptability among adults with ADHD. Some have posited that group CBT may be especially helpful for college students with ADHD in order to facilitate peer normalization and acceptance (Chew et al. 2009).

Two primary feasibility challenges exist in developing interventions for ADHD among college students: (1) cost-effectiveness to accommodate limited budgets for disability support and student mental health services and (2) ease of dissemination and implementation of the treatment by typical providers of support services to college students. Group-based skills-training interventions have shown promising results among adults with ADHD (e.g., Solanto et al. 2010) and may provide a relatively cost-effective and easily disseminated treatment model for treating college students with ADHD.

Summary

The college years are a critical period in the developmental trajectory of young adults. College students with ADHD are at higher risk for maladaptive functioning across several domains, including academic performance, psychological functioning, and social relationships. They tend to have lower GPAs, higher rates of academic probation and dropout, higher depressive symptoms, and lower self-reported quality of life than their non-ADHD peers. For many emerging adults with ADHD, this trajectory of impaired functioning extends into adulthood, where ADHD is associated with greater rates of unemployment and underemployment, economic problems, depression and anxiety, relationship difficulties, and diminished life satisfaction (e.g., Biederman et al. 2006a).

At present, very few published research studies have evaluated the efficacy of psychosocial treatments for ADHD among college students; furthermore, there are no known randomized trials on treatment of any kind for ADHD in college. A recent comprehensive review of ADHD symptomatology among college students highlighted the urgent need for treatment development research with this population. "Given the unique developmental needs and environmental challenges that college students with ADHD face, it is imperative that controlled investigations of pharmacotherapy, psychosocial, and educational interventions be conducted" (Weyandt and DuPaul 2008, p. 316).

Given the dearth of evidence on treatment efficacy with this population, the current review seeks to inform treatment and research by combining evidence from three key domains that inform treatment for college students with ADHD: (1) quantitative and qualitative data on functional impairment associated with ADHD among college students; (2) etiology of ADHD and the developmental context for ADHD among emerging adults, and college students in particular; and (3) treatment outcome research on adolescents and adults with ADHD.

The current review discussed several proposed intervention targets, including: (1) inattentive/disorganized symptoms; (2) college-specific academic skills; (3) daily life functions; (4) emotion regulation; (5) social functioning; and (6) accessing external supports. Further research is needed to evaluate these proposed targets and to develop recommendations for prioritizing targets and selecting specific interventions. Finally, the current review discussed key issues and challenges associated with treatment development research for ADHD among college students, including: (1) effective intervention for emerging adults with still-maturing cognitive control systems located in "high-temptation" contexts; (2) the cyclical demands of the academic schedule; (3) enhancing and sustaining motivation for change; 4) generalizing and maintaining behavior change; and (5) intervention acceptability and feasibility.

For the roughly 5 % of college students with ADHD, there may be great potential for powerful and lasting benefit from effective, evidence-based intervention. Based upon the promising results from initial trials of psychosocial interventions for adults with ADHD, we believe that appropriately adapted interventions for college students will be met with similar success. We hope that the current review provides useful information and guidance to inform much-needed treatment development research in this area.

Conflict of interest None.

Ethical standards The authors have complied with ethical standards of the American Psychology Association and the 1964 Declaration of Helsinki.

References

Allsopp, D. H., Minskoff, E. H., & Bolt, L. (2005). Individualized course-specific strategy instruction for college students with learning disabilities and ADHD: Lessons learned from a model demonstration project. *Learning Disabilities Research & Practice, 20*, 103–118.

Baddeley, A. D., & Hitch, G. (1974). Working memory. In G. H. Bower (Ed.), *The psychology of learning and motivation* (Vol. 8, pp. 47–89). New York: Academic Press.

Barkley, R. A. (1997). Attention-deficit/hyperactivity disorder, self-regulation, and time: Toward a more comprehensive theory. *Journal of Developmental and Behavioral Pediatrics, 18*, 271–279.

Barkley, R. A. (2002). Major life activity and health outcomes associated with attention-deficit/hyperactivity disorder. *Journal of Clinical Psychiatry, 63*, 10–15.

Barkley, R. A. (2006). *Attention-deficit hyperactivity disorder: A handbook for diagnosis and treatment* (3rd ed.). New York: Guilford.

Barkley, R. A., Anastopoulos, A. D., Guevremont, D. C., & Fletcher, K. E. (1992). Adolescents with attention deficit hyperactivity disorder: Mother-adolescent interactions, family beliefs and conflicts, and maternal psychopathology. *Journal of Abnormal Child Psychology, 20*, 263–288.

Barkley, R. A., Edwards, G., Laneri, M., Fletcher, K., & Metevia, L. (2001). Executive functioning, temporal discounting, and sense of time in adolescents with attention deficit hyperactivity disorder (ADHD) and oppositional defiant disorder (ODD). *Journal of Abnormal Child Psychology, 29*, 541–556.

Barkley, R. A., Murphy, K. R., Dupaul, G. I., & Bush, T. (2002). Driving in young adults with attention deficit hyperactivity disorder: Knowledge, performance, adverse outcomes, and the role of executive functioning. *Journal of the International Neuropsychological Society, 8*, 655–672.

Barkley, R. A., Murphy, K. R., & Fischer, M. (2007). *ADHD in adults: What the science says*. New York: Guilford.

Benton, D., Brett, V., & Brain, P. F. (1987). Glucose improves attention and reaction to frustration in children. *Biological Psychology, 24*, 95–100.

Betz, N. E. (2000). Self-efficacy theory as a basis for career assessment. *Journal of Career Assessment, 8*, 205–222.

Biederman, J., Faraone, S. V., Spencer, T. J., Mick, E., Monuteaux, M. C., & Aleardi, M. (2006a). Functional impairments in adults with self-reports of diagnosed ADHD: A controlled study of 1001 adults in the community. *Journal of Clinical Psychiatry, 67*, 524–540.

Biederman, J., Petty, C. R., Fried, R., Doyle, A. E., Spencer, T., Seidman, L. J., et al. (2007). Stability of executive function deficits into young adult years: A prospective longitudinal follow-up study of grown up males with ADHD. *Acta Psychiatrica Scandinavica, 116*, 129–136.

Biederman, J., Petty, C., Fried, R., Fontanella, J., Doyle, A. E., Seidman, L. J., et al. (2006b). Impact of psychometrically defined deficits of executive functioning in adults with attention deficit hyperactivity disorder. *American Journal of Psychiatry, 163*, 1730–1738.

Blase, S. L., Gilbert, A. N., Anastopoulos, A. D., Costello, E. J., Hoyle, R. H., Swartzwelder, H. S., et al. (2009). Self-reported ADHD and adjustment in college: Cross-sectional and longitudinal findings. *Journal of Attention Disorders, 13*, 297–309.

Boonstra, A. M., Oosterlaan, J., Sergeant, J. A., & Buitelaar, J. K. (2005). Executive functioning in adult ADHD: A meta-analytic review. *Psychological Medicine, 35*, 1097–1108.

Britton, B. K., & Tesser, A. (1991). Effects of time-management practices on college grades. *Journal of Educational Psychology, 83*, 405–410.

Brod, M., Johnston, J., Able, S., & Swindle, R. (2006). Validation of the *Adult Attention-Deficit/Hyperactivity Disorder Quality-of-Life Scale* (AAQoL): A disease-specific quality-of-life measure. *Quality of Life Research, 15*, 117–129.

Brod, M., Perwien, A., Adler, L., Spencer, T., & Johnston, J. (2005). Conceptualization and assessment of quality of life for adults with attention-deficit/hyperactivity disorder. *Primary Psychiatry, 12*, 58–64.

Buboltz, W. C., Loveland, J., Jenkins, S. M., Brown, F. C., Soper, B., & Hodges, J. (2006). College student sleep: Relationship to health and academic performance. In M. V. Landow (Ed.), *College students: Mental health and coping strategies* (pp. 1–39). Hauppauge, NY: Nova Science Publishers.

Caldwell, K., Harrison, M., Adams, M., Quin, R. H., & Greeson, J. (2010). Developing mindfulness in college students through movement-based courses: Effects on self-regulatory self-efficacy, mood, stress, and sleep quality. *Journal of American College Health, 58*, 433–442.

Canu, W. H., & Carlson, C. L. (2003). Differences in heterosocial behavior and outcomes of ADHD-symptomatic subtypes in a college sample. *Journal of Attention Disorders, 6*, 123–133.

Carey, K. B., Henson, J. M., Carey, M. P., & Maisto, S. A. (2007). Which heavy drinking college students benefit from a brief motivational intervention? *Journal of Consulting and Clinical Psychology, 75*, 663–669.

Carmody, J., & Baer, R. A. (2008). Relationships between mindfulness practice and levels of mindfulness, medical and psychological symptoms and well-being in a mindfulness-based stress reduction program. *Journal of Behavioral Medicine, 31*, 23–33.

Carroll, B. C., McLaughlin, T. J., & Blake, D. R. (2006). Patterns and knowledge of nonmedical use of stimulants among college students. *Archives of Pediatrics and Adolescent Medicine, 160*, 481–485.

Castellanos, F. X., & Tannock, R. (2002). Neuroscience of attention-deficit/hyperactivity disorder: The search for endophenotypes. *Nature Reviews Neuroscience, 3*, 617–628.

Cauffman, E., & Steinberg, L. (2000). (Im)maturity of judgment in adolescence: Why adolescents may be less culpable than adults. *Behavioral Sciences and The Law, 18*, 741–760.

Chew, B. L., Jensen, S. A., & Rosen, L. A. (2009). College students' attitudes toward their ADHD peers. *Journal of Attention Disorders, 13*, 271–276.

Chronis, A. M., Jones, H. A., & Raggi, V. L. (2006). Evidence-based psychosocial treatments for children and adolescents with attention-deficit/hyperactivity disorder. *Clinical Psychology Review, 26*, 486–502.

Derogatis, L. R. (1975). *The SCL-90-R*. Baltimore, MD: Clinical Psychometric Research.

Derogatis, L. R. (1992). *The Brief Symptom Inventory*. Baltimore, MD: Clinical Psychometric Research.

Dickstein, S. G., Bannon, K., Castellanos, F. X., & Milham, M. P. (2006). The neural correlates of attention deficit hyperactivity disorder: An ALE meta-analysis. *Journal of Child Psychology and Psychiatry and Allied Disciplines, 47*, 1051–1062.

DuPaul, G. J., Weyandt, L. L., O'Dell, S. M., & Varejao, M. (2009). College students with ADHD: Current status and future directions. *Journal of Attention Disorders, 13*, 234–250.

Durmer, J. S., & Dinges, D. F. (2005). Neurocognitive consequences of sleep deprivation. *Seminars in Neurology, 25*, 117–129.

Durston, S., Davidson, M. C., Mulder, M. J., Spicer, J. A., Galvan, A., Tottenham, N., et al. (2007). Neural and behavioral correlates of expectancy violations in attention-deficit hyperactivity disorder. *Journal of Child Psychology and Psychiatry and Allied Disciplines, 48*, 881–889.

Egger, J., Carter, C. M., Graham, P. J., Gumley, D., & Soothill, J. F. (1985). Controlled trial of oligoantigenic treatment in the hyperkinetic syndrome. *Lancet, 1*, 540–545.

Epstein, J. N., Langberg, J. M., Rosen, P. J., Graham, A., Narad, M. E., Antonini, T. N., et al. (2011). Evidence for higher reaction time variability for children with ADHD on a range of cognitive tasks including reward and event rate manipulations. *Neuropsychology, 25*, 427–441.

Ernst, M., Nelson, E. E., Jazbec, S., McClure, E. B., Monk, C. S., Leibenluft, E., et al. (2005). Amygdala and nucleus accumbens in responses to receipt and omission of gains in adults and adolescents. *NeuroImage, 25*, 1279–1291.

Ervin, R. A., DuPaul, G. J., Kern, L., & Friman, P. C. (1998). Classroom-based functional and adjunctive assessments: Proactive approaches to intervention selection for adolescents with attention deficit hyperactivity disorder. *Journal of Applied Behavioral Analysis, 31*, 65–78.

Evans, S. W., Axelrod, J., & Langberg, J. M. (2004). Efficacy of a school-based treatment program for middle school youth with ADHD. *Behavior Modification, 28*, 528–547.

Evans, S. W., Langberg, J. M., Raggi, V. L., Allen, J., & Buvinger, E. C. (2005). Development of a school-based treatment program for middle school youth with ADHD. *Journal of Attention Disorders, 9*, 343–353.

Fafrowicz, M., Oginska, H., Mojsa-Kaja, J., Marek, T., Golonka, K., & Tucholska, K. (2010). Chronic sleep deficit and performance of a sustained attention task: An electrooculography study. *Chronobiology International, 27*, 934–944.

Fan, J., McCandliss, B. D., Sommer, T., Raz, A., & Posner, M. I. (2002). Testing the efficiency and independence of attentional networks. *Journal of Cognitive Neuroscience, 14*, 340–347.

Faraone, S., & Doyle, A. E. (2001). The nature and heritability of attention-deficit/hyperactivity disorder. *Child and Adolescent Psychiatric Clinics of North America, 10*, 299–316.

Farrell, E. F. (2003, Sept. 26, 2002). Paying attention to students who can't. *Chronicle of Higher Education, 50*, A50.

Fedele, D., Hartung, C., Canu, W., & Wilkowski, B. (2010). Potential symptoms of ADHD for emerging adults. *Journal of Psychopathology and Behavioral Assessment, 32*, 385–396.

Fischoff, B. (1992). Risk-taking: A developmental perspective. In J. Yates (Ed.), *Risk-taking behavior* (pp. 133–162). New York: John Wiley.

Frazier, T. W., Youngstrom, E. A., Glutting, J. J., & Watkins, M. W. (2007). ADHD and achievement: Meta-analysis of the child, adolescent, and adult literatures and a concomitant study with college students. *Journal of Learning Disabilities, 40*, 49–65.

Gailliot, M. T., Baumeister, R. F., DeWall, C. N., Maner, J. K., Plant, E. A., Tice, D. M., et al. (2007). Self-control relies on glucose as a limited energy source: Willpower is more than a metaphor. *Journal of Personality and Social Psychology, 92*, 325–336.

Gardner, M., & Steinberg, L. (2005). Peer influence on risk taking, risk preference, and risky decision making in adolescence and adulthood: An experimental study. *Developmental Psychology, 41*, 625–635.

Gau, S. S., Kessler, R. C., Tseng, W. L., Wu, Y. Y., Chiu, Y. N., Yeh, C. B., et al. (2007). Association between sleep problems and symptoms of attention-deficit/hyperactivity disorder in young adults. *Sleep, 30*, 195–201.

Giedd, J. N. (2004). Structural magnetic resonance imaging of the adolescent brain. *Annals of the New York Academy of Sciences, 1021*, 77–85.

Glutting, J., Monaghan, M. C., Adam, W., & Sheslow, D. (2002). Some psychometric properties of a system to measure ADHD among college students: Factor pattern, reliability, and one-year predictive validity. *Measurement and Evaluation in Counseling and Development, 34*, 194–209.

Goldstein, S. (2005). Coaching as a treatment for ADHD. *Journal of Attention Disorders, 9*, 379–381.

Gregg, N., Coleman, C., Stennett, R. B., & Davis, M. (2002). Discourse complexity of college writers with and without disabilities: A multidimensional analysis. *Journal of Learning Disabilities, 35*(23–38), 56.

Grenwald-Mayes, G. (2001). Relationship between current quality of life and family of origin dynamics for college students with attention-deficit/hyperactivity disorder. *Journal of Attention Disorders, 5*, 211–222.

Gropper, R. J., & Tannock, R. (2009). A pilot study of working memory and academic achievement in college students with ADHD. *Journal of Attention Disorders, 12*, 574–581.

Hall, K. M., Irwin, M. M., Bowman, K. A., Frankenberger, W., & Jewett, D. C. (2005). Illicit use of prescribed stimulant medication among college students. *Journal of American College Health, 53*, 167–174.

Halmøy, A., Fasmer, O. B., Gillberg, C., & Haavik, J. (2009). Occupational outcome in adult ADHD: Impact of symptom profile, comorbid psychiatric problems, and treatment: A cross-sectional study of 414 clinically diagnosed adult ADHD patients. *Journal of Attention Disorders, 13*, 175–187.

Halperin, J. M., & Healey, D. M. (2011). The influences of environmental enrichment, cognitive enhancement, and physical exercise on brain development: Can we alter the developmental trajectory of ADHD? *Neuroscience and Biobehavioral Reviews, 35*, 621–634.

Heiligenstein, E., Guenther, G., Levy, A., Savino, F., & Fulwiler, J. (1999). Psychological and academic functioning in college students with attention deficit hyperactivity disorder. *Journal of American College Health, 47*, 181–185.

Heiligenstein, E., Johnston, H. F., & Nielsen, J. K. (1996). Pemoline therapy in college students with attention deficit hyperactivity disorder: A retrospective study. *Journal of American College Health, 45*, 35–39.

Heiligenstein, E., & Keeling, R. P. (1995). Presentation of unrecognized attention deficit hyperactivity disorder in college students. *Journal of American College Health, 43*, 226–228.

Hervey, A. S., Epstein, J. N., & Curry, J. F. (2004). Neuropsychology of adults with attention-deficit/hyperactivity disorder: A meta-analytic review. *Neuropsychology, 18*, 485–503.

Hesslinger, B., Tebartz van Elst, L., Nyberg, E., Dykierek, P., Richter, H., Berner, M., et al. (2002). Psychotherapy of attention deficit hyperactivity disorder in adults: A pilot study using a structured skills training program. *European Archives of Psychiatry and Clinical Neuroscience, 252*, 177–184.

Huizinga, M., Dolan, C. V., & van der Molen, M. W. (2006). Age-related change in executive function: Developmental trends and a latent variable analysis. *Neuropsychologia, 44*, 2017–2036.

Kabat-Zinn, J. (1982). An outpatient program in behavioral medicine for chronic pain patients based on the practice of mindfulness meditation: Theoretical considerations and preliminary results. *General Hospital Psychiatry, 4*, 33–47.

Kern, R., Rasmussen, P. R., Byrd, S. L., & Wittschen, L. K. (1999). Lifestyle, personality, and attention deficit hyperactivity disorder in young adults. *The Journal of Individual Psychology, 55*, 186–199.

Knight, L. A., Rooney, M., & Chronis-Tuscano, A. (2008). Psychosocial treatments for attention-deficit/hyperactivity disorder. *Current Psychiatry Reports, 10*, 412–418.

Knouse, L. E., Cooper-Vince, C., Sprich, S., & Safren, S. A. (2008). Recent developments in the psychosocial treatment of adult ADHD. *Expert Review of Neurotherapeutics, 8*, 1537–1548.

Kubik, J. A. (2010). Efficacy of ADHD coaching for adults with ADHD. *Journal of Attention Disorders, 13*, 442–453.

Lambek, R., Tannock, R., Dalsgaard, S., Trillingsgaard, A., Damm, D., & Thomsen, P. H. (2010). Validating neuropsychological subtypes of ADHD: How do children with and without an executive function deficit differ? *Journal of Child Psychology and Psychiatry and Allied Disciplines, 51*, 895–904.

Lewandowski, L. J., Lovett, B. J., Codding, R. S., & Gordon, M. (2008). Symptoms of ADHD and academic concerns in college students with and without ADHD diagnoses. *Journal of Attention Disorders, 12*, 156–161.

Lijffijt, M., Kenemans, J. L., Verbaten, M. N., & van Engeland, H. (2005). A meta-analytic review of stopping performance in attention-deficit/hyperactivity disorder: Deficient inhibitory motor control? *Journal of Abnormal Psychology, 114*, 216–222.

Linehan, M. M. (1993). *Cognitive-behavioral treatment of borderline personality disorder*. New York: Guilford.

Linterman, I., & Weyandt, L. (2001). Divided attention skills in college students with ADHD: Is it advantageous to have ADHD? *ADHD Report, 9*, 1–10.

Liston, C., Watts, R., Tottenham, N., Davidson, M. C., Niogi, S., Ulug, A. M., et al. (2006). Frontostriatal microstructure modulates efficient recruitment of cognitive control. *Cerebral Cortex, 16*, 553–560.

Loo, S. K., Humphrey, L. A., Tapio, T., Moilanen, I. K., McGough, J. J., McCracken, J. T., et al. (2007). Executive functioning among Finnish adolescents with attention-deficit/hyperactivity disorder. *Journal of the American Academy of Child and Adolescent Psychiatry, 46*, 1594–1604.

Low, K. G., & Gendaszek, A. E. (2002). Illicit use of psychostimulants among college students: A preliminary study. *Psychology, Health & Medicine, 7*, 283–287.

Luman, M., Oosterlaan, J., & Sergeant, J. A. (2005). The impact of reinforcement contingencies on AD/HD: A review and theoretical appraisal. *Clinical Psychology Review, 25*, 183–213.

Makris, N., Biederman, J., Valera, E. M., Bush, G., Kaiser, J., Kennedy, D. N., et al. (2007). Cortical thinning of the attention and executive function networks in adults with attention-deficit/hyperactivity disorder. *Cerebral Cortex, 17*, 1364–1375.

Marco, R., Miranda, A., Schlotz, W., Melia, A., Mulligan, A., Muller, U., et al. (2009). Delay and reward choice in ADHD: An experimental test of the role of delay aversion. *Neuropsychology, 23*, 367–380.

McCabe, S. E., & Boyd, C. J. (2005). Sources of prescription drugs for illicit use. *Addictive Behaviors, 30*, 1342–1350.

McCabe, S. E., Knight, J. R., Teter, C. J., & Wechsler, H. (2005). Non-medical use of prescription stimulants among US college students: Prevalence and correlates from a national survey. *Addiction, 100*, 96–106.

McCleary, L., & Ridley, T. (1999). Parenting adolescents with ADHD: Evaluation of a psychoeducation group. *Patient Education and Counseling, 38*, 3–10.

Meaux, J. B., Green, A., & Broussard, L. (2009). ADHD in the college student: A block in the road. *Journal of Psychiatric and Mental Health Nursing, 16*, 248–256.

Meaux, J. B., Hester, C., Smith, B., & Shoptaw, A. (2006). Stimulant medications: A trade-off? The lived experience of adolescents with ADHD. *Journal for Specialists in Pediatric Nursing, 11*, 214–226.

Medina, J. A., Netto, T. L., Muszkat, M., Medina, A. C., Botter, D., Orbetelli, R., et al. (2010). Exercise impact on sustained attention of ADHD children, methylphenidate effects. *Attention Deficit and Hyperactivity Disorders, 2*, 49–58.

Morris, G. B. (1993). A rational-emotive treatment program with conduct disorder and attention-deficit hyperactivity disorder adolescents. *Journal of Rational-Emotive & Cognitive-Behavior Therapy, 11*, 123–134.

Murphy, K. R., & Barkley, R. A. (1996). Attention deficit hyperactivity disorder adults: Comorbidities and adaptive impairments. *Comprehensive Psychiatry, 37*, 393–401.

Murphy, K. R., Barkley, R. A., & Bush, T. (2001). Executive functioning and olfactory identification in young adults with attention deficit-hyperactivity disorder. *Neuropsychology, 15*, 211–220.

Nigg, J. T., Willcutt, E. G., Doyle, A. E., & Sonuga-Barke, E. J. (2005). Causal heterogeneity in attention-deficit/hyperactivity disorder: Do we need neuropsychologically impaired subtypes? *Biological Psychiatry, 57*, 1224–1230.

Norvilitis, J. M., Sun, L., & Zhang, J. (2010). ADHD symptomatology and adjustment to college in China and the United States. *Journal of Learning Disabilities, 43*, 86–94.

Norwalk, K., Norvilitis, J. M., & MacLean, M. G. (2009). ADHD symptomatology and its relationship to factors associated with college adjustment. *Journal of Attention Disorders, 13*, 251–258.

Pelsser, L. M., Frankena, K., Toorman, J., Savelkoul, H. F., Dubois, A. E., Pereira, R. R., et al. (2011). Effects of a restricted elimination diet on the behaviour of children with attention-deficit hyperactivity disorder (INCA study): A randomised controlled trial. *Lancet, 377*, 494–503.

Philipsen, A., Richter, H., Peters, J., Alm, B., Sobanski, E., Colla, M., et al. (2007). Structured group psychotherapy in adults with attention deficit hyperactivity disorder: Results of an open multicentre study. *The Journal of Nervous and Mental Disease, 195*, 1013–1019.

Prince, J. B. (2006). Pharmacotherapy of attention-deficit hyperactivity disorder in children and adolescents: Update on new stimulant preparations, atomoxetine, and novel treatments. *Child and Adolescent Psychiatric Clinics of North America, 15*, 13–50.

Rabiner, D. L., Anastopoulos, A. D., Costello, J., Hoyle, R. H., & Swartzwelder, H. S. (2008). Adjustment to college in students with ADHD. *Journal of Attention Disorders, 11*, 689–699.

Ramirez, C. A., Rosen, L. A., Deffenbacher, J. L., Hurst, H., Nicoletta, C., Rosencranz, T., et al. (1997). Anger and anger expression in adults with high ADHD symptoms. *Journal of Attention Disorders, 2*, 115–128.

Reaser, A., Prevatt, F., Petscher, Y., & Proctor, B. (2007). The learning and study strategies of college students with ADHD. *Psychology in the Schools, 44*, 627–638.

Richards, T. L., Deffenbacher, J., & Rosen, L. (2002). Driving anger and other driving-related behaviors in high and low ADHD symptom college students. *Journal of Attention Disorders, 6*, 25–38.

Richards, T. L., Rosen, L. A., & Ramirez, C. A. (1999). Psychological functioning differences among college students with confirmed ADHD, ADHD by self-report only, and without ADHD. *Journal of College Student Development, 40*, 299–304.

Rooney, M., Chronis-Tuscano, A., & Yoon, Y. (2012). Substance use in college students with ADHD. *Journal of Attention Disorders, 16*, 221–234.

Rostain, A. L., & Ramsay, J. R. (2006). A combined treatment approach for adults with ADHD: Results of an open study of 43 patients. *Journal of Attention Disorders, 10*, 150–159.

Safren, S. A., Otto, M. W., Sprich, S., Winett, C. L., Wilens, T. E., & Biederman, J. (2005). Cognitive-behavioral therapy for ADHD in medication-treated adults with continued symptoms. *Behaviour Research and Therapy, 43*, 831–842.

Safren, S. A., Sprich, S. E., Cooper-Vince, C., Knouse, L. E., & Lerner, J. A. (2010a). Life impairments in adults with medication-treated ADHD. *Journal of Attention Disorders, 13*, 524–531.

Safren, S. A., Sprich, S., Mimiaga, M. J., Surman, C., Knouse, L., Groves, M., et al. (2010b). Cognitive behavioral therapy vs relaxation with educational support for medication-treated adults with ADHD and persistent symptoms: A randomized controlled trial. *Journal of the American Medical Association, 304*, 875–880.

Sagvolden, T., Johansen, E. B., Aase, H., & Russell, V. A. (2005). A dynamic developmental theory of attention-deficit/hyperactivity disorder (ADHD) predominantly hyperactive/impulsive and combined subtypes. *Behavioral and Brain Sciences, 28*, 397–419. discussion 419–368.

Salakari, A., Virta, M., Gronroos, N., Chydenius, E., Partinen, M., Vataja, R., et al. (2010). Cognitive-behaviorally-oriented group rehabilitation of adults with ADHD: Results of a 6-month follow-up. *Journal of Attention Disorders, 13*, 516–523.

Scheres, A., Milham, M. P., Knutson, B., & Castellanos, F. X. (2007). Ventral striatal hyporesponsiveness during reward anticipation in attention-deficit/hyperactivity disorder. *Biological Psychiatry, 61*, 720–724.

Schulenberg, J. E., Sameroff, A. J., & Cicchetti, D. (2004). The transition to adulthood as a critical juncture in the course of psychopathology and mental health. *Development and Psychopathology, 16*, 799–806.

Schumaker, J. B., Denton, P. H., & Deshler, D. D. (1984). *The paraphrasing strategy*. Lawrence, KS: University of Kansas.

Schwanz, K. A., Palm, L. J., & Brallier, S. A. (2007). Attention problems and hyperactivity as predictors of college grade point average. *Journal of Attention Disorders, 11*, 368–373.

Sergeant, J. A. (2000). The cognitive-energetic model: An empirical approach to attention-deficit hyperactivity disorder. *Neuroscience and Biobehavioral Reviews, 24*, 7–12.

Sergeant, J. A. (2005). Modeling attention-deficit/hyperactivity disorder: A critical appraisal of the cognitive-energetic model. *Biological Psychiatry, 57*, 1248–1255.

Shaw-Zirt, B., Popali-Lehane, L., Chaplin, W., & Bergman, A. (2005). Adjustment, social skills, and self-esteem in college students with symptoms of ADHD. *Journal of Attention Disorders, 8*, 109–120.

Shiels, K., Hawk, L. W., Jr, Reynolds, B., Mazzullo, R. J., Rhodes, J. D., Pelham, W. E., Jr, et al. (2009). Effects of methylphenidate on discounting of delayed rewards in attention deficit/hyperactivity disorder. *Experimental and Clinical Psychopharmacology, 17*, 291–301.

Shifrin, J. G., Proctor, B. E., & Prevatt, F. F. (2010). Work performance differences between college students with and without ADHD. *Journal of Attention Disorders, 13*, 489–496.

Singleton, R. A., Jr, & Wolfson, A. R. (2009). Alcohol consumption, sleep, and academic performance among college students. *Journal of Studies on Alcohol and Drugs, 70*, 355–363.

Smalley, S. L., Loo, S. K., Hale, T. S., Shrestha, A., McGough, J., Flook, L., et al. (2009). Mindfulness and attention deficit hyperactivity disorder. *Journal of Clinical Psychology, 65*, 1087–1098.

Smith, B. H., Molina, B. S. G., & Pelham, W. E. (2002). The clinically meaningful link between alcohol use and attention deficit hyperactivity disorder. *Alcohol Research and Health, 26*, 122–129.

Solanto, M. V., Abikoff, H., Sonuga-Barke, E., Schachar, R., Logan, G. D., Wigal, T., et al. (2001). The ecological validity of delay aversion and response inhibition as measures of impulsivity in AD/HD: A supplement to the NIMH Multimodal Treatment Study of AD/HD. *Journal of Abnormal Child Psychology, 29*, 215–228.

Solanto, M. V., Marks, D. J., Mitchell, K. J., Wasserstein, J., & Kofman, M. D. (2008). Development of a new psychosocial treatment for adult ADHD. *Journal of Attention Disorders, 11*, 728–736.

Solanto, M. V., Marks, D. J., Wasserstein, J., Mitchell, K., Abikoff, H., Alvir, J. M., et al. (2010). Efficacy of meta-cognitive therapy for adult ADHD. *The American Journal of Psychiatry, 167*, 958–968.

Sonuga-Barke, E. J. (2005). Causal models of attention-deficit/hyperactivity disorder: From common simple deficits to multiple developmental pathways. *Biological Psychiatry, 57*, 1231–1238.

Sonuga-Barke, E. J., Bitsakou, P., & Thompson, M. (2010). Beyond the dual pathway model: Evidence for the dissociation of timing, inhibitory, and delay-related impairments in attention-deficit/hyperactivity disorder. *Journal of the American Academy of Child and Adolescent Psychiatry, 49*, 345–355.

Sonuga-Barke, E. J., Dalen, L., & Remington, B. (2003). Do executive deficits and delay aversion make independent contributions to preschool attention-deficit/hyperactivity disorder symptoms? *Journal of the American Academy of Child and Adolescent Psychiatry, 42*, 1335–1342.

Sonuga-Barke, E. J., Houlberg, K., & Hall, M. (1994). When is "impulsiveness" not impulsive? The case of hyperactive children's cognitive style. *Journal of Child Psychology and Psychiatry and Allied Disciplines, 35*, 1247–1253.

Stavro, G. M., Ettenhofer, M. L., & Nigg, J. T. (2007). Executive functions and adaptive functioning in young adult attention-deficit/hyperactivity disorder. *Journal of the International Neuropsychological Society, 13*, 324–334.

Steinberg, L. (2007). Risk taking in adolescence: New perspectives from brain and behavioral science. *Current Directions in Psychological Science, 16*, 55–59.

Steinberg, L. (2010). A dual systems model of adolescent risk-taking. *Developmental Psychobiology, 52*, 216–224.

Steinberg, L., Graham, S., O'Brien, L., Woolard, J., Cauffman, E., & Banich, M. (2009). Age differences in future orientation and delay discounting. *Child Development, 80*, 28–44.

Stevenson, C. S., Whitmont, S., Bornholt, L., Livesey, D., & Stevenson, R. J. (2002). A cognitive remediation programme for adults with attention deficit hyperactivity disorder. *The Australian and New Zealand Journal of Psychiatry, 36*, 610–616.

Strohle, A., Stoy, M., Wrase, J., Schwarzer, S., Schlagenhauf, F., Huss, M., et al. (2008). Reward anticipation and outcomes in adult males with attention-deficit/hyperactivity disorder. *NeuroImage, 39*, 966–972.

Substance Abuse and Mental Health Services Administration. (2007). *Results from the 2006 National Survey on Drug Use and Health: National findings*. Rockville, MD: SAMHSA.

Tang, Y. Y., Ma, Y., Wang, J., Fan, Y., Feng, S., Lu, Q., et al. (2007). Short-term meditation training improves attention and self-regulation. *Proceedings of the National academy of Sciences of the United States of America, 104*, 17152–17156.

Turnock, P., Rosen, L. A., & Kaminski, P. L. (1998). Differences in academic coping strategies of college students who self-report high and low symptoms of attention deficit hyperactivity disorder. *Journal of College Student Development, 39*, 484–493.

Upadhyaya, H. P., Rose, K., Wang, W., O'Rourke, K., Sullivan, B., Deas, D., et al. (2005). Attention-deficit/hyperactivity disorder, medication treatment, and substance use patterns among adolescents and young adults. *Journal of Child and Adolescent Psychopharmacology, 15*, 799–809.

Vaidya, C. J., & Stollstorff, M. (2008). Cognitive neuroscience of attention deficit hyperactivity disorder: Current status and working hypotheses. *Developmental Disabilities Research Reviews, 14*, 261–267.

Van Dongen, H. P., Maislin, G., Mullington, J. M., & Dinges, D. F. (2003). The cumulative cost of additional wakefulness: Dose-response effects on neurobehavioral functions and sleep physiology from chronic sleep restriction and total sleep deprivation. *Sleep, 26*, 117–126.

Virta, M., Salakari, A., Antila, M., Chydenius, E., Partinen, M., Kaski, M., et al. (2010). Short cognitive behavioral therapy and cognitive training for adults with ADHD: A randomized controlled pilot study. *Neuropsychiatric Disease and Treatment, 6*, 443–453.

Virta, M., Vedenpaa, A., Gronroos, N., Chydenius, E., Partinen, M., Vataja, R., et al. (2008). Adults with ADHD benefit from cognitive-behaviorally oriented group rehabilitation: A study of 29 participants. *Journal of Attention Disorders, 12*, 218–226.

Vloet, T. D., Marx, I., Kahraman-Lanzerath, B., Zepf, F. D., Herpertz-Dahlmann, B., & Konrad, K. (2010). Neurocognitive performance in children with ADHD and OCD. *Journal of Abnormal Child Psychology, 38*, 961–969.

Wallis, D., Russell, H. F., & Muenke, M. (2008). Review: Genetics of attention deficit/hyperactivity disorder. *Journal of Pediatric Psychology, 33*, 1085–1099.

Weinstein, C. E., Schulte, A. C., & Palmer, D. R. (2002). *Learning and study strategies inventory–second edition (LASSI-2)*. Clearwater, FL: H & H Publishing.

Weiss, M. D., Gibbins, C., Goodman, D. W., Hodgkins, P. S., Landgraf, J. M., & Faraone, S. V. (2010). Moderators and mediators of symptoms and quality of life outcomes in an open-label study of adults treated for attention-deficit/hyperactivity disorder. *Journal of Clinical Psychiatry, 71*, 381–390.

Weyandt, L. L., & DuPaul, G. J. (2008). ADHD in college students: Developmental findings. *Developmental Disabilities Research Reviews, 14*, 311–319.

Weyandt, L. L., Linterman, I., & Rice, J. A. (1995). Reported prevalence of attention deficits in a general sample of college students. *Journal of Psychopathology and Behavioral Assessment, 17*, 293–304.

Weyandt, L. L., Mitzlaff, L., & Thomas, L. (2002). The relationship between intelligence and performance on the Test of Variables of Attention (TOVA). *Journal of Learning Disabilities, 35*, 114–120.

Weyandt, L. L., Rice, J. A., Linterman, I., Mitzlaff, L., & Emert, E. (1998). Neuropsychological performance of a sample of adults with ADHD, developmental reading disorder, and controls. *Developmental Neuropsychology, 14*, 643–656.

Wigal, S. B. (2009). Efficacy and safety limitations of attention-deficit hyperactivity disorder pharmacotherapy in children and adults. *CNS Drugs, 23*(Suppl 1), 21–31.

Wiggins, D., Singh, K., Getz, H. G., & Hutchins, D. E. (1999). Effects of brief group intervention for adults with attention deficit/hyperactivity disorder. *Journal of Mental Health Counseling, 21*, 82–92.

Wilens, T. E., Biederman, J., & Spencer, T. J. (1998). Pharmacotherapy of attention deficit hyperactivity disorder in adults. *CNS Drugs, 9*, 347–356.

Wilens, T. E., Spencer, T. J., & Biederman, J. (2002). A review of the pharmacotherapy of adults with attention-deficit/hyperactivity disorder. *Journal of Attention Disorders, 5*, 189–202.

Wilens, T. E., Vitulano, M., Upadhyaya, H., Adamson, J., Sawtelle, R., Utzinger, L., et al. (2008). Cigarette smoking associated with attention deficit hyperactivity disorder. *Journal of Pediatrics, 153*, 414–419.

Willcutt, E. G., Doyle, A. E., Nigg, J. T., Faraone, S. V., & Pennington, B. F. (2005). Validity of the executive function theory of attention-deficit/hyperactivity disorder: A meta-analytic review. *Biological Psychiatry, 57*, 1336–1346.

Wolf, L. E. (2001). College students with ADHD and other hidden disabilities: Outcomes and interventions. *Annals of the New York Academy of Sciences, 931*, 385–395.

Woodward, L. J., Fergusson, D. M., & Horwood, L. J. (2000). Driving outcomes of young people with attentional difficulties in adolescence. *Journal of the American Academy of Child and Adolescent Psychiatry, 39*, 627–634.

Xu, J. (2010). Predicting homework time management at the secondary school level: A multilevel analysis. *Learning and Individual Differences, 20*, 34–39.

Zwart, L. M., & Kallemeyn, L. M. (2001). Peer-based coaching for college students with ADHD and learning disabilities. *Journal of Postsecondary Education and Disability, 15*, 1–15.

Zylowska, L., Ackerman, D. L., Yang, M. H., Futrell, J. L., Horton, N. L., Hale, T. S., et al. (2008). Mindfulness meditation training in adults and adolescents with ADHD: A feasibility study. *Journal of Attention Disorders, 11*, 737–746.